



Fat, muscle, and anti-obesity medications in cardiovascular disease prevention

Muhammad Shahzeb Khan ^{1,2,3,*}, Muhammad Hamza Dawood ^{4,5},
Yehuda Handelsman ⁶, Stefan D. Anker ⁷, Andrew J. Stewart Coats ⁸,
Jennifer B. Green ⁹, and Javed Butler ^{1,10,*}

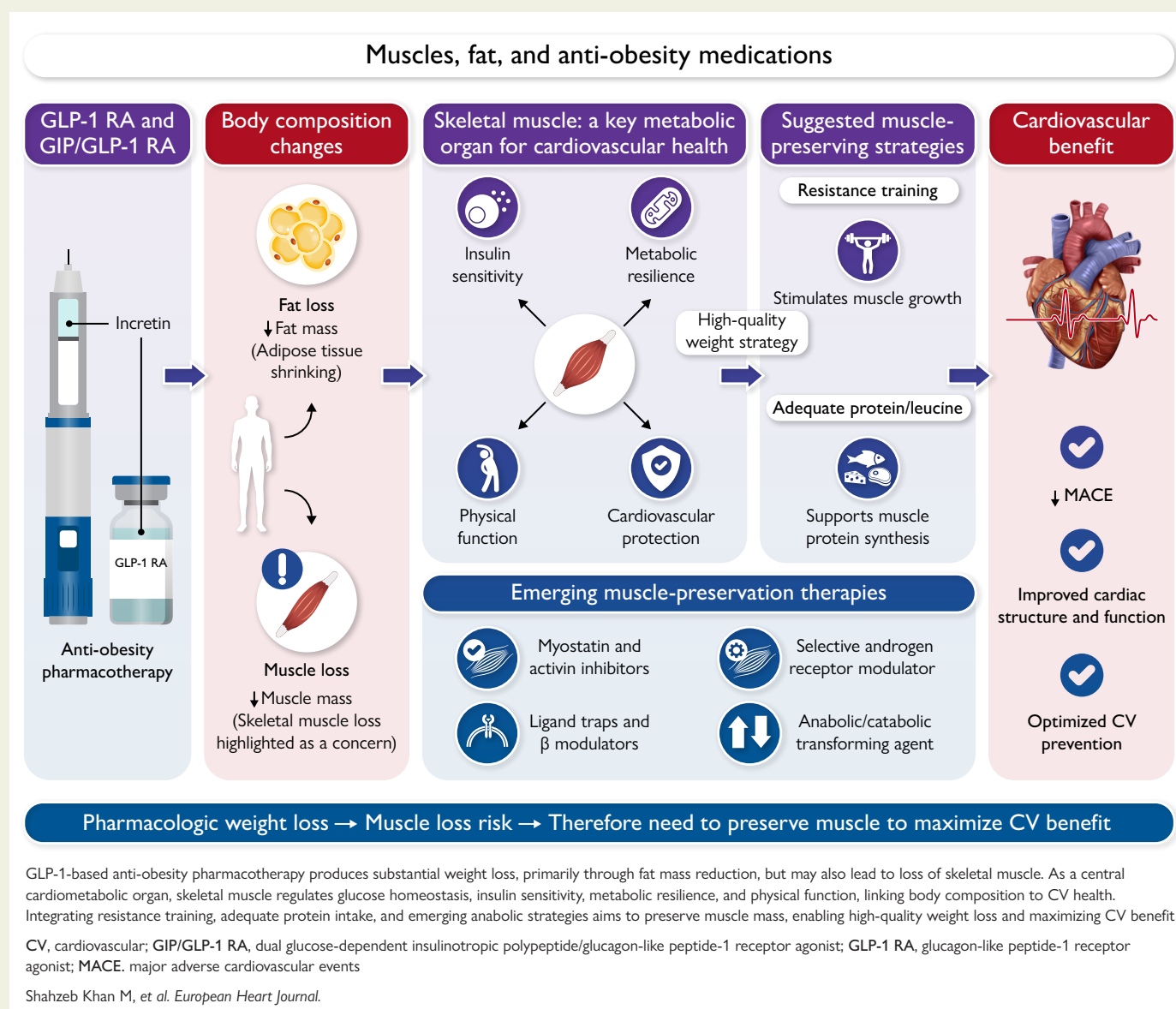
¹Baylor Scott and White Research Institute, Baylor Scott and White Health, 3434 Live Oak Street, Dallas, TX 75204 USA; ²Division of Cardiology, Baylor Scott and White: The Heart Hospital Plano, Plano, TX, USA; ³Division of Cardiology, Baylor College of Medicine, Temple, TX, USA; ⁴Division of Cardiology, Abbasi Shaheed Hospital, Karachi, Pakistan; ⁵Division of Pharmacology, United Medical and Dental College, Karachi, Pakistan; ⁶Division of Cardiology, Metabolic Institute of America, Tarzana, CA, USA; ⁷German Heart Center Charité, German Centre for Cardiovascular Research (DZHK), Partner Site Berlin, Charité Universitätsmedizin, Berlin, Germany; ⁸Division of Cardiology, The Heart Research Institute, University of Sydney, NSW, Australia; ⁹Duke Clinical Research Institute, Duke University School of Medicine, Durham, NC, USA; and ¹⁰Division of Cardiology, University of Mississippi Medical Center, Jackson, MS, USA

Received 16 September 2025; revised 2 January 2026; accepted 1 March 2026; online publish-ahead-of-print 31 March 2026

* Corresponding authors. Tel: 214-820-2687, Email: Shahzeb.Khan@bswhealth.org (M.S.K.); Email: Javed.Butler@BSWHealth.org (J.B.)

© The Author(s) 2026. Published by Oxford University Press on behalf of the European Society of Cardiology. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

Graphical Abstract



Incretin-based antiobesity therapies are often accompanied by reductions in lean body mass, including skeletal muscle, raising concerns regarding sarcopenia, frailty, and diminished metabolic resilience that may attenuate long-term cardiovascular risk reduction. Achieving 'high-quality weight loss' requires preservation of skeletal muscle through resistance training and optimized nutritional strategies. Emerging therapies targeting the myostatin–activin signalling pathway, ligand traps, and selective androgen receptor modulators may enhance muscle quality and act synergistically with incretin-based therapies to optimize body composition and support durable cardiovascular prevention.

Abstract

The rapid expansion of anti-obesity treatments with glucagon-like peptide-1 receptor agonists has redefined weight management. A consistent component of this weight loss, however, involves not only fat mass but also lean body mass, including skeletal muscle. This raises concerns regarding sarcopenia, frailty, and metabolic resilience that may attenuate long-term cardiovascular risk reduction. Muscle loss with these drugs is multifactorial, related to caloric restriction, anabolic resistance, and hormonal shifts. Emerging agents targeting the myostatin/activin pathway, ligand traps, and selective androgen receptor modulators may increase muscle quality and have synergistic benefit with incretin-based therapies. Resistance training is currently the suggested strategy for preserving skeletal muscle and functional capacity during pharmacologic weight loss, while adjunctive strategies such as optimized protein intake and

nutraceuticals may further mitigate muscle catabolism. A paradigm shift is needed in obesity treatment away from total weight loss towards high-quality weight loss that preserves or enhances muscle mass, optimizing body composition and supporting durable cardiovascular risk reduction. Future research should study lean mass preservation as a treatment goal, redefine trial endpoints, and validate emerging combination interventions for optimal body composition. This manuscript reviews the evidence on muscle loss with pharmacologic weight loss therapies, its mechanistic underpinnings, explores emerging agents designed to preserve lean tissue, and outlines strategies to optimize body composition in the context of cardiovascular prevention.

Keywords

Obesity • Semaglutide • Tirzepatide • GLP-1 receptor agonist • Lean body mass • Sarcopenia • Myostatin/activin receptor inhibitor • Myostatin/activin pathway • Selective androgen receptor modulator • Muscle preservation • Weight loss therapy • Body composition • Cardiovascular prevention

Introduction

According to the World Health Organization, an estimated 890 million adults globally are obese and over 40% or 2.5 billion individuals have excess adiposity,^{1–4} a figure projected to surpass 50% by 2035.⁵ This carries major health implications as obesity is a well-established cardiovascular risk factor predisposing to and worsening numerous cardiovascular diseases,⁶ as emphasized in a recent clinical consensus statement.⁷ Obesity imposes substantial economic burden, with global costs projected to reach \$4.32 trillion annually by 2035 from healthcare expenditures and diminished workforce productivity.⁸ Management of obesity has evolved from relying solely on lifestyle interventions to incorporating pharmacologic therapies,^{9,10} ranging from appetite suppressants to potent anorectic effects that modulate appetite pathways.^{11–13} Earlier therapies like orlistat, phentermine/topiramate, and naltrexone/bupropion demonstrated modest weight reductions,^{11,14} but their use was limited by safety concerns.^{15,16}

Newer agents such as glucagon-like peptide-1 receptor agonists (GLP-1 RAs), e.g. semaglutide and dual glucose-dependent insulinotropic polypeptide (GIP)/GLP-1 RA, e.g. tirzepatide, deliver superior weight loss of ~15%–25%.^{17–21} A considerable proportion of this weight loss involves lean tissue, including muscle mass, raising concerns about reduced strength, frailty, and impaired metabolic health, that may lessen the benefits associated with weight loss and impact cardiovascular health. Preserving lean mass is important for sustaining metabolic resilience, functional capacity, and achieving durable cardiovascular protection. The implications of treatment-induced changes in body composition remain underrecognized. This review synthesizes current evidence regarding muscle mass loss associated with incretin-based therapies, explore emerging drugs designed to preserve muscle mass, and examines adjunctive interventions to mitigate muscle loss, emphasizing cardiovascular outcomes (*Graphical Abstract*).

Muscle mass loss with obesity pharmacotherapy

Incretin-based drugs induce weight loss through appetite suppression, reduced energy intake,²² satiety,²³ and metabolic modulation.^{24–27} A considerable fraction of weight loss with these drugs comprises of lean mass, including skeletal muscle loss.^{28,29} Muscle loss may attenuate short-term benefit and raise cardiovascular risk in the long run by reducing glucose disposal, lowering peak exercise oxygen consumption (VO₂), and

metabolic flexibility.^{29–31} The loss of muscle mass is related to a catabolic state, impaired anabolic signalling, and reduced muscle protein synthesis.^{32,33} A principal mechanism is the development of a negative nitrogen balance due to caloric restriction,³⁴ which promotes a catabolic state wherein endogenous protein stores are mobilized to meet metabolic demands.³⁵ This accelerates skeletal muscle proteolysis to provide amino acids for gluconeogenesis and other physiological functions, shifting muscle protein balance towards degradation.³⁶ Inadequate protein intake impairs postprandial stimulation of muscle protein synthesis, with leucine deficiency serving as a critical factor in the suppression of mechanistic target of rapamycin complex 1 (mTORC1) signalling and anabolic resistance.³⁷

Caloric restriction and reduced nutrient intake disrupts key anabolic signalling for maintaining muscle mass.³⁸ These agents are also associated with reductions in circulating insulin and insulin-like growth factor-1 (IGF-1), both of which activate phosphatidylinositol 3-kinase (PI3K)-Akt/mTORC1 pathway responsible for muscle protein synthesis.^{39,40} Attenuation of insulin/IGF-1 signalling reduces mTORC1 activity and impairs protein translation, diminishing protein synthesis.⁴¹ Reduced Akt phosphorylation permits nuclear translocation of Forkhead box-O (FoxO) transcription factors, induces the expression of atrophy-related genes such as muscle RING finger 1 (MuRF-1) and Atrogin-1-key E3 ubiquitin ligases that facilitate skeletal muscle proteolysis.^{42,43} Elevated myostatin levels may further suppress mTOR signalling and exacerbate muscle catabolism⁴⁴ (*Figure 1*). These alterations that favours lean mass attenuate improvements in insulin sensitivity, endothelial function, and vascular health, central to cardiovascular prevention.^{45–47}

Weight regain typically occurs after discontinuation of therapy, with a subset of individuals regaining nearly all lost weight over time, predominantly as fat mass. These patterns may have cardiovascular implications,⁴⁸ as well as visceral fat regain drives insulin resistance, dyslipidaemia, inflammation, and hypertension, diminishing the cardiometabolic benefits in the long term.^{49,50}

Magnitude of lean mass loss

Incretin-based therapies report lean body mass changes broadly, encompassing various tissues, including skeletal muscle, without explicitly differentiating muscle mass. Consequently, lean body mass and muscle mass are often used interchangeably, despite the distinctions between the two (*Table 1*).^{51–65} Liraglutide has demonstrated modest effects on lean mass. In

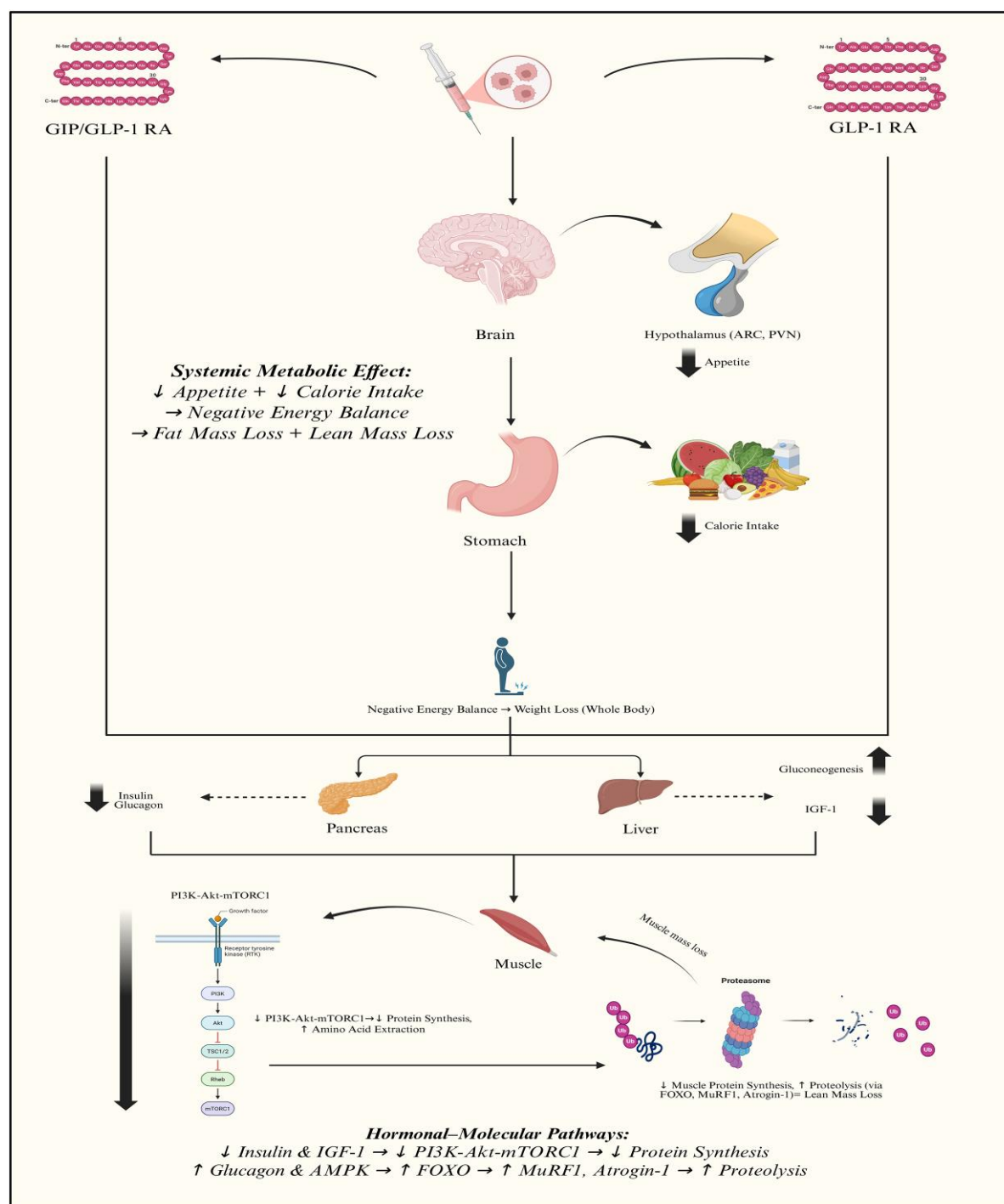


Figure 1 Mechanistic pathways of lean mass loss with incretin-based therapies. This schematic illustrates the integrated neurohormonal and metabolic mechanisms through which glucose-dependent insulinotropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1) receptor agonists induce weight loss, while also highlighting their unintended impact on skeletal muscle. Central effects on the hypothalamus particularly the arcuate nucleus (ARC) and paraventricular nucleus (PVN)—lead to appetite suppression and reduced caloric intake, resulting in a systemic negative energy balance. Simultaneously, peripheral mechanisms including decreased circulating insulin and insulin-like growth factor 1 (IGF-1), along with increased glucagon levels and activation of 5' adenosine monophosphate-activated protein kinase (AMPK), converge to suppress the phosphoinositide 3-kinase–protein kinase B–mechanistic target of rapamycin complex 1 (PI3K–Akt–mTORC1) signalling pathway. This suppression reduces muscle protein synthesis and amino acid uptake. Moreover, activation of forkhead box O (FOXO) transcription factors lead to upregulation of muscle-specific ubiquitin ligases—muscle RING-finger protein-1 (MuRF-1) and muscle atrophy F-box (Atrogin-1) enhancing protein degradation via the ubiquitin–proteasome system. Collectively, these pathways result in the concurrent loss of fat mass and lean muscle mass during pharmacologic weight reduction

Table 1 Magnitude of lean/muscle and fat loss across key obesity pharmacotherapy studies

Study (Year)	Population	Study design	Drug and dose	Duration	Method of body composition	Lean/muscle mass change	Fat mass change
Santini et al. (2023) ⁵¹	n = 54 adults with obesity or overweight with ≥ comorbidity	Prospective observational cohort	Liraglutide 3.0 mg daily	10 months	DEXA	−3.3 kg (−5.9%)	−10.4 kg (−19.1%)
Silver et al. (2023) ⁵²	Adults with obesity and prediabetes	RCT	Liraglutide 1.8 mg daily	14 weeks	DEXA	−0.8 kg (−1.5%)	−1.5 kg (−3.0%)
Yu et al. (2021) ⁵³	63 obese adults with T2D	RCT	Liraglutide 1.8 mg daily, SC	12 weeks	BIA	No significant change (≈ 0 kg)	−1.97 kg (not % reported)
Ghanim et al. (2020) ⁵⁴	84 overweight/obese insulin-treated adults with T1D	RCT	Liraglutide 1.8 mg daily	26 weeks	DEXA + MRI	~0 kg (no significant change)	−3.5 kg (−9%)
Wilding et al., STEP1 sub-study (2021) ⁵⁵	140 adults with overweight/obesity	RCT	Semaglutide 2.4 mg SC, weekly	68 weeks	DEXA	−3.3 kg (−9.7%)	−9.5 kg (−19.3%)
McCrimmon et al., SUSTAIN 8 sub-study (2020) ⁵⁶	Subset of adults with T2D on metformin	RCT	Semaglutide 1.0 mg SC, weekly	52 weeks	DEXA	−2.3 kg (−4.5%)	−3.4 kg (−10.2%)
Rodríguez Jiménez et al. (2024) ⁵⁷	Adults with T2D and obesity (n = 55 SC, n = 33 oral)	Quasi-experimental retrospective study	Subcutaneous Semaglutide 0.25–1.0 mg weekly vs Oral Semaglutide 7–14 mg/day	24 weeks	BIA	SC: −1.1 kg (−3.3%) Oral: −0.5 kg (−1.7%)	SC: −8.5 kg (−16.8%) Oral: −8.0 kg (−20.1%)
Uchiyama et al. (2023) ⁵⁸	n = 25 Japanese adults with T2D	24-week retrospective BIA study	Oral semaglutide, titrated to 14 mg daily	24 weeks	BIA	−0.5 kg (−1%)	−2.8 kg (−10%)
Volpe et al. (2023) ⁵⁹	n = 32 adults with T2D and obesity	Prospective observational study	Oral semaglutide 7–14 mg/day	6 months	Phase-sensitive octopolar SMF-BIA	+0.3 kg (+1.5%)	−5.2 kg (−17.6%)
Chun et al. (2025) ⁶⁰	n = 94 adults	Retrospective real-world cohort	Weekly compounded semaglutide (0.25 → 2.4 mg)	3 months	DEXA/BIA	−1.43 kg (−1.43%) LBM and −0.88 kg skeletal muscle	−2.67 kg total fat mass
Garvey et al., REDEFINE 1 (2025) ⁶¹	n ≈ 3417 adults without T2D, BMI ≥ 30 or ≥ 27 kg/m ² + comorbidity	Phase IIIa randomized, double-blind, placebo- and active-controlled	CagriSema (semaglutide 2.4 mg + cagrilintide 2.4 mg weekly)	68 weeks	DEXA (subgroup n = 252)	Lean soft-tissue ↓ ~14.4% (~33% of total loss)	Fat mass ↓ ~35.7% (~67%)

Continued

Table 1 Continued

Study (Year)	Population	Study design	Drug and dose	Duration	Method of body composition	Lean/muscle mass change	Fat mass change
Look et al., SURMOUNT-1 sub-study (2025) ⁶²	160 adults with overweight/obesity (mean baseline weight 102.5 kg; BMI 38 kg/m ²)	RCT	Tirzepatide 5/10/15 mg weekly (pooled doses)	72 weeks	DEXA	−5.5 kg (25.0%)	−16.3 kg (75.0%)
Zinn et al. (2025) ⁶³	68-year-old male, BMI 31.2 kg/m ² (n = 1)	Case report with serial DEXA scans	Tirzepatide (dose not specified)	61 weeks	DEXA	−4.5 kg (−15.5%)	−8.6 kg (−25.6%)
Adamidis et al. (2025) ⁶⁴	115 obese/overweight adults (67.5% female; median age: 48 years)	Prospective observational cohort	Tirzepatide 2.5–5 mg weekly	30 days	BIA (Akern BIA 101)	−0.7 kg (−1.16%)	−3.6 kg (−11.2%)
Sattar et al. (2025) ⁶⁵	n = 190 adults with T2D (tirzepatide pooled group)	post hoc analysis of RCT (SURPASS-3 MRI)	Tirzepatide (5, 10, 15 mg/week pooled)	52 weeks	MRI (AMRA Researcher)	−0.6 to −0.7 L (−5.5% to −7.3%) Z-score Δ: −0.19 to −0.29	Bodyweight ↓ −9.0% to −11.9% Muscle fat infiltration ↓ −3.5% to −5.5%

BIA, Bioelectrical impedance analysis; BMI, body mass index; DEXA, dual-energy X-ray absorptiometry; LBM, lean body mass; MRI, magnetic resonance imaging; SC, subcutaneous; SMF-BIA, segmental multifrequency bioelectrical impedance analysis; T1D, type 1 diabetes; T2D, type 2 diabetes; RCT, randomized controlled trial.

a real-world cohort of adults with obesity receiving 3.0 mg daily liraglutide for 10 months, Santini *et al.* reported a lean mass loss of -5.9% and a fat mass reduction of -19.1% .⁵¹ In a 14-week trial among adults with obesity and prediabetes, Silver *et al.* observed a -1.5% decrease in lean mass and a -3.0% fat mass loss.⁵² Liraglutide in patients with type 2 diabetes have reported more attenuated lean mass changes. Yu *et al.*, reported no significant change in lean mass after 12 weeks of 1.8 mg/day liraglutide, although fat mass declined by 1.97 kg. Small sample size, short follow-up and use of bioimpedance in this study warrant cautious interpretation.⁵³ Ghanim *et al.* found no significant lean mass change after 26 weeks of liraglutide 1.8 mg in patients with type 1 diabetes, despite a -9% fat mass reduction.⁵⁴

Semaglutide exhibits more pronounced weight loss and lean mass loss. In the STEP-1 sub-study, weekly 2.4 mg semaglutide over 68 weeks led to 9.7% lean mass loss and 19.3% fat mass loss.⁵⁵ In a 52-week SUSTAIN8 sub-study, 1.0 mg semaglutide showed a 4.5% lean and a 10.2% fat mass loss, reflecting a lean-to-fat loss ratio consistent with higher-dose findings.⁵⁶ In another study comparing subcutaneous (SC) and oral semaglutide, both formulations resulted in similar fat mass reductions (SC -8.5 kg, oral -8.0 kg) with relatively modest declines in lean mass (SC 1.1 kg [3.3%], oral 0.5 kg [1.7%]).⁵⁷ Other studies with oral semaglutide suggest a more favourable lean mass preservation profile. Uchiyama *et al.* reported a 1% lean mass with a 10% fat mass reduction over 24 weeks in Japanese adults with type 2 diabetes.⁵⁸ In contrast, Volpe *et al.* observed a 1.5% gain in lean mass and a 17.6% reduction in fat mass in a 6-month prospective study.⁵⁹ A real-world cohort treated with compounded semaglutide demonstrated a 1.43% decline in lean mass and 0.88 kg skeletal muscle loss, accompanied by a 2.67 kg fat mass reduction over 3 months.⁶⁰ Garvey *et al.* reported a 14.4% reduction in lean soft tissue and a 35.7% fat mass loss with CagriSema (semaglutide 2.4 mg + cagrilintide 2.4 mg weekly) in the REDEFINE 1 trial.⁶¹

Tirzepatide, a dual GIP/GLP-1 RA, has shown potent effects on both fat and lean mass. In the SURMOUNT-1 sub-study, pooled data from 5/10/15 mg weekly doses over 72 weeks demonstrated a 25.0% lean and 75.0% fat mass reduction of total weight loss.⁶² A 61-week case report by Zinn *et al.* documented a 15.5% lean mass and 25.6% fat mass reduction in a 68-year-old male on tirzepatide, showing a $\sim 2:1$ fat-to-lean loss ratio.⁶³ Adamidis *et al.* showed early effects over 30 days and found a 1.16% lean and 11.2% fat mass loss in a real-world cohort.⁶⁴ Sattar *et al.* in a *post hoc* analysis of the SURPASS-3 MRI substudy, reported a 5.5% to 7.3% decline in thigh muscle volume and a 3.5% to 5.5% reduction in muscle fat infiltration with tirzepatide, alongside a 9.0% to 11.9% reduction in bodyweight.⁶⁵

Across trials, dual GIP/GLP-1 RAs with tirzepatide is associated with greater total weight loss and larger absolute fat mass reduction than GLP-1 RAs. However, the relative contribution of lean mass to total weight loss remains comparable. In STEP-1, semaglutide 2.4 mg resulted in $\sim 9.7\%$ lean mass loss alongside $\sim 19.3\%$ fat mass loss over 68 weeks, whereas in SURMOUNT-1, tirzepatide achieved greater overall weight loss, with approximately 25% of weight reduction attributable to lean tissue and 75% to fat mass. These findings suggest that dual GIP/GLP-1 receptor agonism confers augmented fat mass loss without a corresponding attenuation in lean tissue loss relative to GLP-1 receptor agonism alone.

Cardiovascular effects

Skeletal muscle is the principal site of insulin-mediated glucose disposal and a key determinant of cardiorespiratory fitness ($\text{VO}_{2\text{peak}}$).⁶⁶ Disproportionate lean-mass loss with anti-obesity drugs is thought to blunt the expected cardiometabolic improvements achieved with preferential fat reduction.^{67,68} Consequences include attenuated improvements in insulin sensitivity, atherogenic lipoproteins, blood pressure, endothelial function, and systemic inflammation.⁶⁶ Mechanistically, reduced myokine signalling and impaired Akt-endothelial nitric oxide (eNOS) cross-talk can promote endothelial dysfunction and arterial stiffness, while diminished skeletal-muscle 'sink' capacity favours ectopic lipid deposition and dyslipidaemia.^{69–71} These mechanisms provide biological plausibility that preservation or augmentation of skeletal muscle during pharmacologic weight loss could further enhance cardiovascular risk reduction. Incretin-based agents confer cardiovascular benefit through both weight-dependent and weight-independent mechanisms. Preferential loss of visceral and ectopic adiposity improves insulin sensitivity, lowers blood pressure, and favourably modifies atherogenic lipoproteins, thereby attenuating major drivers of atherosclerotic cardiovascular disease. Beyond weight reduction, these agents exert direct vascular and myocardial effects, including enhanced endothelial nitric oxide bioavailability, reduced oxidative stress and inflammation, and improved arterial compliance. They also promote natriuresis, contributing to blood pressure lowering and favourable ventricular loading. In the myocardium, incretin signalling improves glucose utilization and energetic efficiency and may reduce lipotoxicity and ischaemia-reperfusion injury. Incretins improve postprandial lipoprotein metabolism and exert systemic anti-inflammatory and anti-atherogenic effects that may slow plaque progression. This highlights that overall cardiovascular balance remains favourable with newer anti-obesity pharmacotherapy by amplifying improvements in insulin sensitivity, endothelial function, lipid handling, blood pressure regulation, and systemic inflammation.⁷²

Large outcome trials and guideline confirm these benefits with GLP-1 RAs reducing major adverse cardiovascular events (MACE) across diverse high-risk populations,^{73,74} such as type 2 diabetes.⁷⁵ The LEADER trial showed that liraglutide reduced the risk of MACE by 13% compared with placebo in patients with type 2 diabetes and high cardiovascular risk.⁷⁶ The REWIND trial demonstrated a significant 12% reduction in MACE with dulaglutide in patients with and without established cardiovascular disease.⁷⁷ Pooled analyses of SUSTAIN-6 and PIONEER-6 indicate that semaglutide reduced the risk of MACE by approximately 24% compared with placebo in subjects with type 2 diabetes at high cardiovascular risk (hazard ratio [HR] 0.76).⁷⁸ In the SELECT trial, semaglutide 2.4 mg significantly lowered the risk of the primary composite endpoint (cardiovascular death, non-fatal myocardial infarction, or non-fatal stroke) by $\sim 20\%$ compared with placebo in patients with overweight or obesity and established cardiovascular disease without diabetes (6.5% vs 8.0%; HR 0.80; 95% confidence interval [CI] 0.72–0.90; $P < .001$).⁷⁴ In heart failure with preserved ejection fraction, STEP-HFpEF and STEP-HFpEF-DM semaglutide showed improvements in symptoms and exercise capacity in patients with obesity and cardiometabolic risk, 13.7 vs 6.4-point improvements in Kansas City Cardiomyopathy Questionnaire clinical

summary score and ~ 12.7 vs -1.6 m change in 6-min walk distance compared with placebo at 52 weeks, alongside greater weight loss.^{79–82} For dual GIP/GLP-1 RAs, data from the SUMMIT trial show that tirzepatide reduced the composite of cardiovascular death or worsening heart failure events by $\sim 38\%$ (9.9% vs 15.3%; HR 0.62, 95% CI 0.41–0.95; $P = .026$), with greater improvements in health status and exercise capacity and greater weight loss ($\sim 15.7\%$ vs $\sim 2.2\%$ at 52 weeks).⁸³ In the SURPASS-CVOT trial of adults with type 2 diabetes and established atherosclerotic cardiovascular disease, $\sim 13\,000$ participants were randomized to tirzepatide or dulaglutide.⁸⁴ Over a median follow-up of approximately 4 years, the primary composite outcome of cardiovascular death, myocardial infarction, or stroke occurred in 801 of 6586 participants (12.2%) in the tirzepatide group and in 862 of 6579 participants (13.1%) in the dulaglutide group (HR 0.92; 95.3% CI 0.83–1.01; $P = .003$ for non-inferiority; $P = .09$ for superiority), demonstrating non-inferiority of tirzepatide relative to dulaglutide but no statistically significant superiority. However, the lean mass loss raises concerns regarding the attenuation of cardiovascular benefit in the short term and progressively increasing the risk in the long run, highlighting the need for focused research on strategies to optimize skeletal muscle preservation and body composition.

Clinical implications

Lean mass loss during pharmacologic weight reduction has direct clinical and cardiovascular implications for long-term risk modification and functional capacity. GLP-1 RA and GIP/GLP-1 RA achieve substantial weight loss,^{17–21} with 20%–30% typically represented by lean tissue.⁸⁵ Variability in lean mass loss across trials may reflect differences in protein intake and resistance training which are often poorly reported. It is important to differentiate total lean mass loss which commonly includes reductions in glycogen stores, organ mass, and water, from actual skeletal muscle depletion.^{86,87} Most studies indicate that the lean mass loss identified by imaging, e.g. dual-energy X-ray absorptiometry (DEXA), bioelectrical impedance analysis, or magnetic resonance imaging (MRI) may not correlate directly with decreased muscle mass, strength or physical function, as reductions in hepatic glycogen and intracellular water may artificially inflate estimates of muscle protein catabolism.^{88–90} Thus, changes in lean mass alone may not reflect clinically meaningful outcomes, with muscle strength and functional capacity being of greatest relevance when interpreting the net clinical impact of weight loss. Clinically, modest lean mass declines are frequently offset by large cardiometabolic gains, improved insulin sensitivity, reductions in visceral adiposity, favourable shifts in atherogenic lipoproteins, lower blood pressure, and attenuated inflammatory tone yielding a net benefit for cardiovascular prevention.⁹¹ In vulnerable patient groups such as older adults, individuals with sarcopenia, or those prone to frailty, even small declines in muscle may impair functional independence, blunt vascular benefits, and increase fall risk, complicating long-term management.^{92,93} Proactive monitoring of body composition (DEXA, MRI, computed tomography [CT], ultrasound), functional status, muscle function, and cardiometabolic markers may be monitored in clinical care, with early prioritization of resistance

training and optimized protein intake as the primary, evidence-based strategies to preserve skeletal muscle and functional capacity, complemented by adjunctive therapies.^{35,94–96}

Activin–myostatin inhibitors

Myostatin/activin signalling pathway is a principal negative regulator of skeletal muscle growth. Myostatin, also known as growth differentiation factor-8 (GDF-8), is a cytokine within the transforming growth factor-beta (TGF- β) superfamily, secreted by skeletal muscle.⁹⁷ It binds to activin type II receptors (ActRIIA and ActRIIB) on muscle fibres, and recruits type I receptors (ALK4/5) and initiates SMAD2/3 phosphorylation.⁹⁸ The phosphorylated SMAD2/3 associates with SMAD4 and translocates to the nucleus, where it represses transcriptional genes involved in muscle protein synthesis and satellite cell proliferation.^{99,100} Parallel signalling pathways inhibit the AKT/mTOR axis and activate FOXO transcription factors, leading to upregulation of atrophy genes such as MuRF1 and atrogin-1, which drive ubiquitin-proteasome-mediated protein degradation.⁹⁹ Related ligands, including activin A, activin B, and GDF-11, share similar signalling and collectively act to limit muscle hypertrophy.^{100–102} Blocking this pathway is shown to cause muscle hypertrophy (Figure 2). Pharmacologic inhibitors of the myostatin/activin pathway have shown promise in preserving or enhancing muscle mass by disrupting catabolic signalling and restoring anabolic activity during pharmacologic weight loss (Table 2).^{103–123}

Bimagrumab, a fully human monoclonal antibody targeting activin type-II receptors (ActRIIA and ActRIIB), inhibits ligands including myostatin, activin A/B, and GDF-11, thereby promoting skeletal muscle anabolism and concurrently reducing adipose tissue mass through antagonism of the activin–myostatin signalling.¹²⁴ A phase 2 trial in patients with type 2 diabetes and obesity over 48 weeks reported reduced fat mass by 20.5%, increased lean mass by 3.6%, and yielded a net weight loss of 6.5%.¹⁰⁴ Another phase 2 trial reported comparable findings.¹⁰⁵ Subsequent studies have extended this evaluation to patients without diabetes and obesity. The BELIEVE trial is currently ongoing and has reported interim findings reported that at 48 weeks, bimagrumab monotherapy reduced total body fat mass by 8.6 kg with a 0.4 kg increase in lean mass, while the bimagrumab + semaglutide combination produced the greatest effect with a 15.6 kg reduction in fat mass and only a small 1.3 kg decrease in lean mass, supporting a more fat-selective pattern of weight loss compared with incretin-based therapies where a meaningful proportion of weight reduction can reflect lean tissue loss.^{106,107} An ongoing combination study is exploring the synergistic potential of bimagrumab with tirzepatide in inducing greater fat loss while preserving or enhancing muscle mass.¹⁰⁸ Bimagrumab requires monthly intravenous administration and is well tolerated, however, long-term safety, especially related to mineral metabolism, is under investigation.¹²⁵

Apitegromab (SRK-439) is a highly selective inhibitors of myostatin, targeting the precursor form of GDF-8 without binding activin A or GDF-11.¹²⁶ This narrower specificity is designed to preserve skeletal muscle while minimizing off-target effects on vasculature, bone, or reproductive signalling. In the EMBRAZE Phase 2 trial, apitegromab (10 mg/kg) was evaluated in adults with overweight or obesity receiving tirzepatide over 24 weeks. Apitegromab preserved 54.9% of lean mass compared to

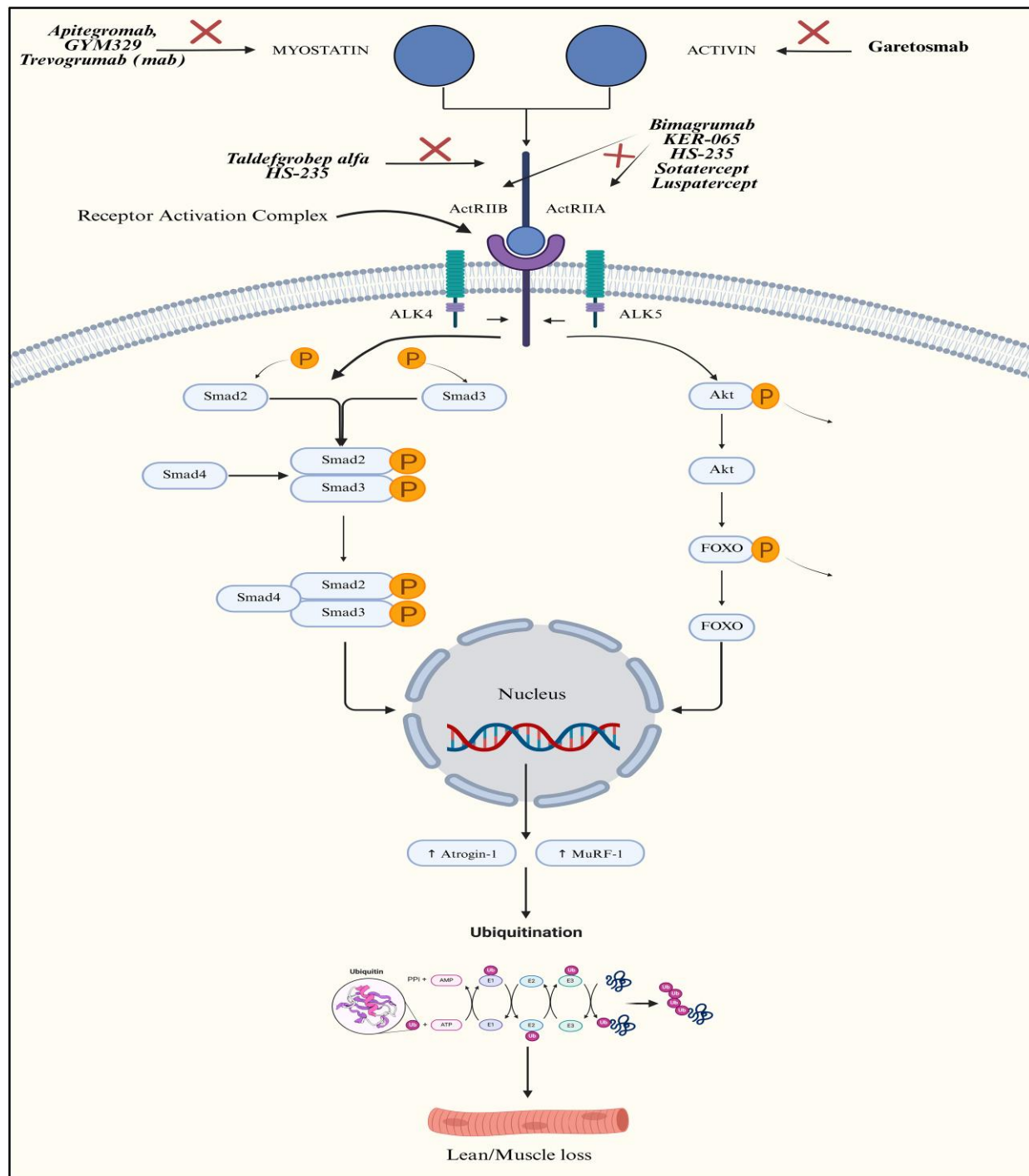


Figure 2 Mechanism of myostatin/activin signalling and target of pharmacologic inhibition. This schematic illustrates the molecular pathway through which myostatin and activin, members of the transforming growth factor-beta (TGF- β) superfamily, signal via activin type II receptors—activin receptor type IIA (ActRIIA) and activin receptor type IIB (ActRIIB)—activating activin receptor-like kinase 4 and 5 (ALK4/ALK5) co-receptors to phosphorylate SMAD family members 2 and 3 (Smad2/3). The Smad2/3/4 complex (including SMAD4) translocates to the nucleus, upregulating muscle-specific ubiquitin ligases such as muscle atrophy F-box (Atrogin-1) and muscle RING-finger protein-1 (MuRF-1), thereby promoting proteasomal degradation of muscle proteins and leading to muscle atrophy. Concurrently, inhibition of the protein kinase B/forkhead box O (Akt/FOXO) signalling axis results in increased FOXO nuclear translocation and transcriptional activation of the same catabolic ligases. Therapeutic agents such as bimagrumab, KER-065, and sotercept inhibit ligand–receptor interactions at various checkpoints, thereby preventing downstream signalling and subsequent muscle loss. Ligand-neutralizing monoclonal antibodies—including apitegromab, garetosmab, and trevogrumab—and receptor blockers such as taldefgrobep alfa further disrupt the myostatin/activin-mediated catabolic cascade

Table 2 Summary of clinical and preclinical studies evaluating myostatin/activin pathway inhibitors and related agents for effects on body composition in obesity

Trial/Study	Phase	Population	Intervention	Duration	Key outcomes
Heymsfield et al. (NCT03005288) ¹⁰⁴	2	T2DM + obesity (BMI 28–40 kg/m ²)	Bimagrumab vs placebo	48 weeks	↓20.5% FM; ↑3.6% LM; ↓6.5% BW
Attie et al. (NCT05616013) ¹⁰⁵	2	Overweight/obesity and T2DM	Bimagrumab vs placebo	48 weeks	↓22.2% FM; ↑2.5% LM; ↓7.5% BW
Versanis Bio/Eli Lilly (BELIEVE; NCT05616013) ^{106,107}	2 (ongoing)	Adults with overweight/obesity (BMI ≥30, or ≥27 kg/m ² with comorbidities); non-diabetic (HbA1c < 6.5%)	Bimagrumab ± semaglutide vs placebo	48 weeks	Monotherapy: ↓8.6 kg total fat mass (DXA); 10.4 kg lean mass (DXA); ↓5.23% body fat; VAT –0.4 kg; trunk fat –5.44 kg; ≥10% BW loss: 39.3%. Combination: ↓15.6 kg total fat mass (DXA); ↓1.3 kg lean mass (DXA); ↓9.30% body fat; VAT –0.7 kg; trunk fat –9.19 kg; ≥10% BW loss: 79.2%, ↓22% FM.
Eli Lilly (NCT06643728) ¹⁰⁸	2	Obesity, overweight, non-diabetic	Bimagrumab + tirzepatide and placebo combo	24–48 weeks	Active recruitment
EMBRAZE ¹⁰⁹	2 (ongoing)	Obese/overweight adults (BMI >27 kg/m ²) on GLP-1 or dual GIP/GLP-1 RA (tirzepatide or semaglutide)	Apitegromab 10 mg/kg IV q4w + GLP-1 or dual GIP/GLP-1 RA vs placebo + GLP-1 or dual GIP/GLP-1 RA	24 weeks	LM preservation: apitegromab preserved +4.2 lbs lean mass (54.9% preservation; P = .001) vs tirzepatide alone. Fat loss: –18.8 lbs with apitegromab + tirzepatide vs –17.7 lbs with tirzepatide alone. Total weight loss: –12.3% with apitegromab vs –13.4% with tirzepatide alone. Body composition: higher-quality weight loss (85% fat/15% LM) vs tirzepatide alone (70% fat/30% LM). Safety: Generally well tolerated, consistent with prior studies.
Fulham et al., Preclinical ¹¹⁰	Preclinical	DIO mice on semaglutide (0.04 mg/kg daily)	SRK-439 (0.3–10 mg/kg SC weekly) vs anti-ActRII vs control (all with semaglutide)	4 weeks	SRK-439 dose-dependently attenuated GLP-1 RA-induced LM loss (e.g. at 10 mg: –4.2% lean vs –14.1% in control); superior muscle preservation vs anti-ActRII at matched doses. FM not reported in this study.
Chugai/Roche Phase ¹¹¹	I	Adults with overweight or obesity (± T2DM)	GYM329 SC 4 week vs placebo	1–2 months	Initiated in 2024; results not yet reported
RESILIENT (Biohaven) ¹¹²	3	SMA type 2 and 3 patients on background nusinersen or risdiplam	Taldefgrobep alfa vs placebo	48 weeks	No significant improvement in primary functional endpoints (e.g. HFMSE); however, muscle volume and quality preserved.
Biohaven (Obesity program; preclinical/early clinical) ¹¹³	1–2 (ongoing)	Obesity-related (non-SMA), under evaluation	Taldefgrobep alfa (monotherapy or combo)	N/A	Exploring potential for muscle mass preservation during pharmacologic weight loss.

Continued

Table 2 Continued

Trial/Study	Phase	Population	Intervention	Duration	Key outcomes
COURAGE (Regeneron) ^{114,115}	2b (interim)	Obesity (BMI ≥30 kg/m ²) adults	Semaglutide alone vs + low-dose trevogrumab vs + high-dose trevogrumab vs + high-dose trevogrumab + garetosmab	26 weeks	LM: −7.9 lb (34.5%) mono vs −3.7 lb (50.8%) preserved low-dose vs −4.2 lb (51.3%) high-dose vs −2.0 lb (80.9%) triplet. FM: −15.3 lb (66.3%) mono vs −16.9 lb (78.1%) low-dose vs −18.9 lb (76.3%) high-dose vs −25.4 lb (84.4%) triplet (27.3% ↑ vs mono). BW: −23.0 lb (10.4%) mono vs −21.6 lb (9.9%) low-dose vs −24.8 lb (11.3%) high-dose vs −30.0 lb (13.2%) triplet ($P < .05$).
Pistilli et al. ¹¹⁶	Preclinical (mdx mouse)	mdx mice (DMD model)	Recombinant soluble ActRIIB (RAP-011) at 1 and 10 mg/kg	12 weeks	At 10 mg/kg: ↑ LM, ↑ overall BM, ↑ gastrocnemius, quadriceps, tibialis anterior ($P < .0001$). Muscle force and function improved; dose-dependent response, little benefit at 1 mg/kg
Attie et al. (NCT01432717) ¹¹⁷	Phase 1	Healthy postmenopausal women ($n = 48$)	ACE-031 (0.02–3 mg/kg, single SC dose) vs placebo	29 days	↑3.3% total LM; ↑5.1% thigh muscle volume; ↓FM; favourable shifts in bone/fat biomarkers; generally well-tolerated
Cadena et al. ¹¹⁸	Preclinical	C57BL/6 male mice (8-week-old)	ACE-031 (soluble ActRIIB receptor) vs vehicle	28 days	↑16% BW; ↑33%–46% in muscle mass (soleus, plantaris, gastrocnemius, EDL); ↑22%–57% fiber CSA in type I and II fibers; no change in fiber-type proportion; hypertrophy independent of fiber type
Nathan et al. ¹¹⁹	Preclinical	Male C57BL/6 mice (12-week-old)	RKER-065 (0.3–20 mg/kg QW) vs vehicle and ActRIIB-Fc	6 weeks	↑ BW (up to ~30%), ↑ LM (up to ~20%), ↑ Muscle weights, ↑ Grip strength, ↑ Bone volume fraction.
Keros Therapeutics (KER-065) ^{120,121}	1	Healthy adult volunteers	Multiple ascending doses of KER-065 vs placebo	~85 days (MAD period)	↑ muscle mass and BMD; ↓ FM consistent with activin inhibition
Schang et al. ¹²²	Preclinical	DIO mice	HS235 with or without GLP-1 RA	~6–8 weeks	HS235 alone: ↓ FM, ↑ LM and muscle hypertrophy, HS235 + incretin mimetic combo: synergistic fat loss and prevention of LM loss seen with incretins alone.
35Pharma (HS235) ¹²³	Preclinical	DIO mice	HS235 + caloric restriction vs HS235 + tirzepatide	~6–8 weeks	HS235 + caloric restriction: weight loss exclusively from fat, Tirzepatide alone: −46% FM; HS235 + tirzepatide: −64% FM.

BMD, bone mineral density; BMI, body mass index; BW, body weight; CSA, cross-sectional area; DIO, diet-induced obese; DMD, Duchenne muscular dystrophy; EDL, extensor digitorum longus; FM, fat mass; GIP, glucose-dependent insulintropic polypeptide; GLP-1, glucagon-like peptide-1; GLP-1 RA, glucagon-like peptide-1 receptor agonist; HVA1c, glycated haemoglobin; HFSME, Hammersmith Functional Motor Scale Expanded; IV, intravenous; LM, lean mass; MAD, multiple ascending dose; SC, subcutaneous; SMA, spinal muscle atrophy; T2DM, type 2 diabetes mellitus; DXA, dual-energy X-ray absorptiometry.

tirzepatide alone (+4.2 lbs) despite similar total weight loss (−12.3% vs −13.4%), and led to greater fat mass reduction (−18.8 vs −17.7 lbs), with 85% of weight loss derived from fat mass compared to 70% with tirzepatide alone.¹⁰⁹

GYM329 is a Fc-engineered monoclonal antibody targeting latent myostatin, preventing its proteolytic activation to mature GDF-8. By sparing activin A and GDF-11 signalling, it improves specificity, and its high-affinity Fc domain enables FcγRIIb engagement and FcRn-mediated recycling for sustained clearance of circulating myostatin. GYM329 and its surrogates induced skeletal muscle hypertrophy and strength in animal models without adverse systemic toxicity.¹²⁷ The resulting hypertrophy is associated with improved glycaemic control in animal models, likely due to enhanced energy expenditure and insulin sensitivity.¹²⁸ A phase I trial is initiated in overweight and obese adults, including individuals with or without type 2 diabetes.¹¹¹

Taldefgrobep alfa is a recombinant fusion protein that neutralizes myostatin, with potential weak interactions with other ActRII ligands.¹²⁹ Although taldefgrobep did not achieve motor function improvement in RESILIENT spinal muscle atrophy trial, it produced notable anabolic effects on body composition, including reduced fat mass, increased lean mass, and enhanced bone mineral density.¹¹² These data prompted a phase 2 trial planned aiming to assess its efficacy in preserving both muscle and bone mass during weight reduction.¹¹³

Trevogrumab is a myostatin-neutralizing monoclonal antibody being studied with garetosmab, an activin A-neutralizing antibody, to achieve dual ligand blockade of ActRII signalling.¹³⁰ While trevogrumab alone showed modest effects in prior studies, preclinical models demonstrate that co-inhibition of myostatin and activin A leads to better muscle hypertrophy compared to myostatin inhibition alone.¹⁰³ The ongoing COURAGE trial combines trevogrumab (±garetosmab) with semaglutide in adults with obesity.¹¹⁴ Interim results show that semaglutide alone resulted in a lean mass loss of 7.9 lb (34.5% of total weight loss), whereas the addition of low- or high-dose trevogrumab reduced lean mass loss to 3.7 lb (50.8% preserved) and 4.2 lb (51.3% preserved), respectively. The combination of high-dose trevogrumab and garetosmab further preserved lean mass to 2.0 lb loss (80.9% preserved). Fat mass loss increased from 15.3 lb (66.3%) with semaglutide alone to 16.9 lb (78.1%), 18.9 lb (76.3%), and 25.4 lb (84.4%) with the low-dose, high-dose, and triplet combinations, respectively.¹¹⁵

Ligand traps and related TGF-β modulators

Sotatercept (ACE-011) and luspatercept (ACE-536) are ligand trap fusion proteins that modulate the translational potential of TGF-β superfamily. Sotatercept, an ActRIIA-Fc fusion, was developed to enhance bone and muscle mass by neutralizing activins and GDF-11.^{131,132} Preclinical study with its analogue (RAP-011) in mouse model of Duchenne muscular dystrophy (mdx) mice treated with soluble ActRIIB (10 mg/kg for 12 weeks) showed significant muscle mass and strength.¹¹⁶ However, it also enhanced erythropoiesis, leading to approvals for beta-thalassemia-related anaemia. It was also approved for pulmonary arterial hypertension.¹³³ Muscle-targeted development was discontinued due to effects on elevated haemoglobin.¹³⁴

Luspatercept, a modified ActRIIB-Fc fusion with refined ligand specificity, is approved for anaemia in beta-thalassemia and myelodysplastic syndromes.¹³⁵ While it preserves bone mass in preclinical models, its impact on muscle remains unreported.¹¹⁷

KER-065 is a ActRII ligand trap designed to selectively bind myostatin and activin A, while sparing bone-promoting ligands. Engineered for high-affinity interaction with catabolic TGF-β superfamily ligands, KER-065 show preclinical efficacy in increasing skeletal muscle mass function, and preventing bone loss in murine models of atrophy.^{118,119} In a Phase 1 study in healthy adults, KER-065 was well tolerated with biomarker evidence of activin pathway inhibition and improvements in body composition over 12 weeks.¹²⁰ KER-065 will be tested in Duchenne muscular dystrophy, with obesity as a secondary indication.¹²¹

HS235 is a next-generation Fc-fused ligand trap designed to selectively inhibit catabolic TGF-β superfamily members, such as activins and myostatin, while preserving anabolic BMP signalling through high-affinity binding.¹²² Preclinical studies in diet-induced obese mice demonstrate that HS235, either alone or in combination with GLP-1 RA, reduces fat mass and enhances lean mass and muscle hypertrophy.¹²² Combined with tirzepatide, HS235 showed synergistic effects, amplifying fat loss (−64%) compared to tirzepatide alone (−46%) and preventing lean mass loss.¹²³

Selective androgen receptor modulators

Selective androgen receptor modulators (SARM) bind and activate androgen receptors (ARs) in skeletal muscle and bone, eliciting anabolic effects while sparing other androgen-sensitive tissues such as the prostate and sebaceous glands.¹³⁶ This specificity is achieved through differential AR conformational changes, selective recruitment of co-activators, and avoidance of AR-driven gene transcription in off-target organs.¹³⁷ In myocytes, SARM stimulate protein synthesis via enhanced mTOR signalling, increase satellite cell proliferation, and inhibit muscle proteolysis through suppression of ubiquitin-proteasome pathways.¹³⁸ In contrast to testosterone or synthetic anabolic steroids, SARM avoid aromatization to oestrogens and 5α-reductase to dihydrotestosterone (DHT), thereby minimizing risks of gynecomastia, virilization, and prostate hypertrophy¹³⁹ (Figure 3). These properties render SARM attractive for combating muscle atrophy in conditions of caloric deficit, sarcopenia, or drug induced weight loss (Table 3).^{140–148}

Enobosarm (Ostarine, MK-2866, GTx-024) is an oral compound with anabolic effects. Studies in rodents show that enobosarm increases lean mass, preserves bone density, and enhances physical function and avoid off-target androgenic effects.¹⁴⁰ In five trials involving 968 adults, including those with and without cancer-related muscle wasting, enobosarm increased lean mass (~1.3–1.5 kg at 3 mg) and improved muscle strength.¹⁴¹ In QUALITY, enobosarm (3 or 6 mg) was evaluated in 168 older adults with obesity receiving semaglutide 2.4 mg weekly. The 3 mg dose preserved over 99% of lean mass representing ~71% relative reduction in muscle loss vs placebo despite similar total weight loss (~4.5 kg). Enobosarm recipients experienced a 27% greater reduction in fat mass, with only 9.4% of weight loss derived from lean mass vs 31.9% in placebo, with a 55% reduction in proportion of individuals exhibiting ≥10% loss in stair-climb power (19% vs 43%).^{142,143} In the

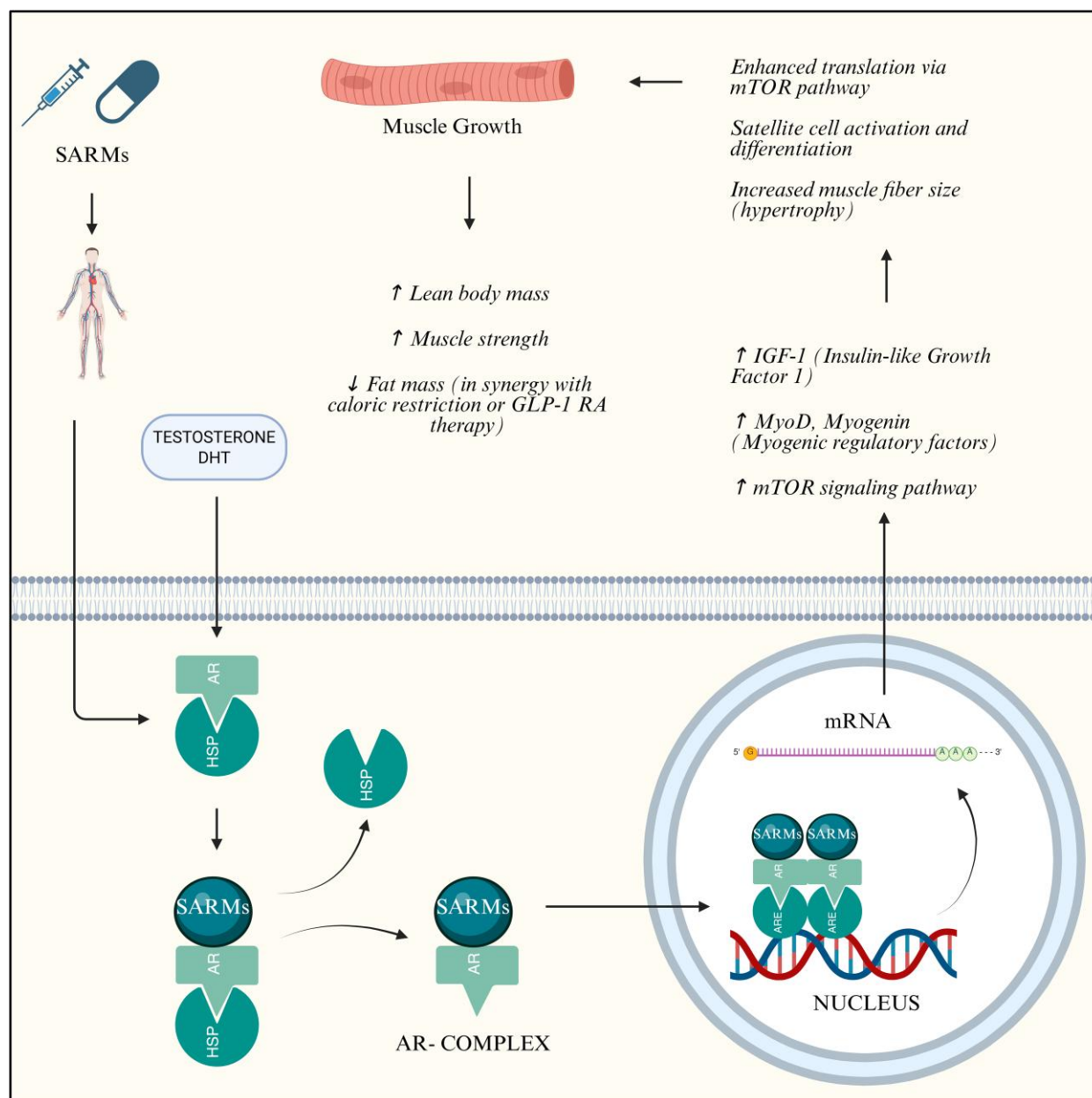


Figure 3 Mechanism of selective androgen receptor modulators to preserve muscle mass during weight loss. This illustration depicts the anabolic signalling cascade initiated by SARMs, which selectively bind to the androgen receptor (AR) and displace heat shock proteins (HSP), forming a SARM-AR complex that translocates to the nucleus. Upon binding to androgen response elements (AREs), this complex stimulates transcription of genes involved in muscle hypertrophy. Downstream effects include upregulation of insulin-like growth factor 1 (IGF-1), MyoD, and myogenin, activation of the mTOR signalling pathway, and enhancement of muscle protein synthesis. These mechanisms collectively lead to increased lean body mass, improved muscle strength, and reduced fat mass particularly when combined with caloric restriction or glucagon-like peptide-1 (GLP-1) receptor agonist therapy

QUALITY extension study, enobosarm (3 or 6 mg) was evaluated in older adults with obesity who discontinued semaglutide after initial weight loss; the 3 mg dose reduced body weight regain by 46% compared to placebo (1.41% vs 2.57% regain) and prevented fat regain (−0.27% fat mass change vs +2.27%), with 100% of weight regained composed of lean mass. Enobosarm 3 mg led to a 58% greater reduction in fat mass (−9.5% vs −6.0%) and preserved lean mass (+2.70% vs −0.94%), while the 6 mg dose achieved a 93% greater fat mass reduction (−11.6%).^{142,143}

LGD-4033 (Ligandrol, VK5211) preserves lean mass during calorie deficits. In a phase I trial of 76 healthy men, daily doses of 0.1–1.0 mg LGD-4033 over 21 days resulted in a dose-dependent increase in lean mass (~1.2 kg at 1 mg/day), with no significant change in fat mass, while maintaining favourable safety and reversible hormonal and lipid effects.¹⁴⁴ Over 12 weeks among hip-fracture patients, LGD-4033 exhibited dose-dependent lean mass gains of +4.8%, +7.2%, and +9.1% at doses of 0.5 mg, 1 mg, and 2 mg respectively.¹⁴⁵

Table 3 Summary of clinical and preclinical studies evaluating selective androgen receptor modulators for effects on body composition in obesity

Trial/study	Phase	Population	Intervention	Duration	Key outcomes
Gao et al. ¹⁴⁰	Preclinical	Castrated male rats (12 weeks post-orchidectomy)	5-4 (3 or 10 mg/kg) vs DHT (3 mg/kg) vs vehicle	8 weeks	↑ Soleus and levator ani muscle mass and strength (restored to intact levels); ↑ LBM
Dalton et al. (Study G200501) ¹⁴¹	2	Males >60 and postmenopausal women	Enobosarm 3 mg vs placebo	12 weeks	↑1.25 kg LBM ($P < .001$); ↑SCP ($P = .049$); ↓0.32 kg fat mass (2%–5%) ($P = .049$)
Merck CSR (Study 003) ¹⁴¹	2	Sarcopenic postmenopausal women	Enobosarm 3 mg vs placebo	12 weeks	↑1.54 kg LBM ($P < .001$); ↑bilateral leg press +21.86 lbs vs +1.5 lbs (placebo); Fat mass not collected
Dobs et al. (Study G200502) ¹⁴¹	2b	Cancer-associated muscle wasting	Enobosarm 3 mg vs placebo	16 weeks	↑1.3 kg LBM ($P = .046$); ↑SCP +16.8 W ($P = .001$); ↓0.76 kg fat mass ($P = .086$)
Study G300504 ¹⁴¹	3	Lung cancer receiving cisplatin + taxane chemotherapy	Enobosarm 3 mg vs placebo	21 weeks	↑0.8 kg LBM Day 84 ($P < .001$); ↑5.17% SCP vs –1.27% placebo ($P = .0147$); Fat mass not collected
Study G300505 ¹⁴¹	3	Lung cancer receiving cisplatin + non-taxane chemotherapy	Enobosarm 3 mg vs placebo	21 weeks	↑0.73 kg LBM Day 84 and ↑0.67 kg Day 147 ($P = .013$); SCP NS; Fat mass not collected
Veru Inc. (Phase 2b QUALITY) ^{142,143}	2b	Older adults (≥60 years) with sarcopenic obesity or overweight on semaglutide	Enobosarm 3 mg or 6 mg + semaglutide vs placebo + semaglutide	16 weeks	Preserved lean mass: 71% less LBM loss vs placebo (1.2% vs 4.1%; $P = .002$); 3 mg dose showed >99% relative preservation ($P < .001$); Greater fat loss (~10.9% vs 8.6%, $P = .096$) Improved body composition: similar total weight loss but higher fat.
Veru Inc. (Phase 2b QUALITY Maintenance Extension) ^{142,143}	2b	Older adults (≥60 years) with sarcopenic obesity or overweight previously on semaglutide	Enobosarm 3 mg or 6 mg monotherapy vs placebo after semaglutide discontinuation	12 weeks maintenance period (total study duration 28 weeks)	Weight regain after semaglutide discontinuation: Placebo regained 2.57% body weight (5.06 lbs, 43% of prior weight loss). Enobosarm 3 mg reduced regain to 1.41% (2.73 lbs, 46% reduction, $P = .038$), while 6 mg showed 2.87% regain (5.29 lbs). Fat regain: Placebo regained +2.27% fat mass (1.22 lbs), while 3 mg enobosarm lost –0.27% fat mass (–0.43 lbs) and 6 mg lost –0.50% (–0.20 lbs); thus both doses prevented fat regain (0% fat regained vs 28% in placebo). Lean mass regain: Placebo regained +3.18% lean mass (72% of weight regained); 3 mg preserved +2.68% lean mass and 6 mg preserved +5.18% (both 100% lean mass preservation). Fat mass loss over full 28 weeks: Placebo lost –6.0% fat mass; 3 mg enobosarm lost –9.5% (58% greater fat loss, $P = .085$), 6 mg lost –11.6% (93% greater fat loss, $P = .008$).

Continued

Table 3 Continued

Trial/study	Phase	Population	Intervention	Duration	Key outcomes
Basaria et al. ¹⁴⁴	1	Healthy men aged 21–50 years	LGD-4033 (0.1, 0.3, 1.0 mg daily) vs placebo	21 days	Lean mass preservation over full 28 weeks: Placebo lost –0.94% lean mass, while 3 mg preserved +2.70% ($P = .001$) and 6 mg preserved +2.36% ($P = .004$). Safety: Both doses well tolerated; no significant GI side effects, no drug-induced liver injury, no increases in obstructive sleep apnoea. Mild, transient liver enzyme elevations and small decreases in eGFR occurred only in 6 mg group, likely related to increased muscle mass. • ↑ LBM dose-dependently (max +1.21 kg at 1.0 mg, $P = .047$ vs placebo)
Viking Therapeutics ¹⁴⁵	2	Patients recovering from non-elective hip fracture surgery	VK5211 (multiple doses, including 2.0 mg) vs placebo	12 weeks	↑ Total LBM (placebo-adjusted +4.7%–8.3%) ↑ Appendicular LBM (placebo-adjusted +6.1%–10.2%) ↑ LBM less head (placebo-adjusted +4.8%–9.1%) ↓ Fat mass (–2.2% to –6.2%, sig. at 2.0 mg) ↑ Body weight and 6MWT (non-significant trend) No drug-related SAEs reported
Miller et al. ¹⁴⁶	Preclinical	Castrated and intact male rats; male cynomolgus monkeys	RAD140 (0.01–30 mg/kg PO)	11–28 days	↑ Levator ani muscle mass at doses ≥ 0.03 mg/kg in rats (equivalent to intact control at 0.3 mg/kg)
Neil et al. ¹⁴⁷	1b	Healthy older men and postmenopausal women	GSK2881078 (multiple escalating doses) vs placebo	Up to 53 days	LBM in a dose-dependent manner. Women showed greater response at lower doses No SAEs.
Mohan et al. ¹⁴⁸ (NCT03359473)	2A	Men and postmenopausal women with COPD (FEV ₁ 30%–65%, SPPB 3–11)	GSK2881078 (2.0 mg men, 1.0 mg women) vs placebo + home exercise	13 weeks	↑ LBM in both sexes (quantified increase not specified)

6MWT, 6-min walk test; COPD, chronic obstructive pulmonary disease; DHT, dihydrotestosterone; eGFR, estimated glomerular filtration rate; FEV₁, forced expiratory volume in 1s; GI, gastrointestinal; LBM, lean body mass; PO, per os; SAE, serious adverse event; SCP, stair-climb power; SPPB, Short Physical Performance Battery.

Table 4 Adjunctive strategies to counter muscle loss in weight loss therapy

Strategy	Effect on muscle/fat	Evidence/efficacy	Integration with Pharma
Resistance training (2–3×/week)	Stimulates muscle protein synthesis; 1 strength; preferentially preserves/builds lean mass during weight loss.	Robust evidence: preserves lean mass in diet-induced loss; in GLP-1 trials, exercise + drug superior to drug alone for body comp.	Strongly recommended for all patients on GLP-1 RAs; low risk, high benefit.
Adequate protein intake (≥1.0–1.2 g/kg)	Provides substrate for muscle maintenance; prevents excessive muscle catabolism.	High-protein diets reduce % lean mass lost; e.g. lean loss halved vs normal protein in some studies. Semaglutide review suggests ensure protein to mitigate lean loss.	Emphasize protein foods; use shakes/supplements if appetite limited. Monitor albumin, prealbumin as surrogates for nutritional status.
Essential amino acids/HMB	Leucine triggers mTOR; HMB attenuates breakdown.	Mixed evidence: HMB preserved lean in some cachexia trials; less proven in obesity but mechanistically sound.	Consider for older patients or those with poor protein intake; relatively low cost.
Creatine supplementation	Improves high-intensity exercise capacity; may increase lean mass (via water/infrastructure).	Shown to aid muscle retention in calorie-restricted athletes (small studies). Generally increases strength when combined with RT.	Safe for most; can be added for those engaging in RT to maximize gains. Ensure good hydration.
Aerobic exercise (cardio)	Increases calorie expenditure mostly from fat; improves cardiovascular fitness; minor direct muscle stimulus (endurance).	Helps target fat for fuel. Combined with RT, yields best functional outcomes. On its own, may not prevent lean loss, but aids overall fat loss and fitness.	Include moderate aerobic (e.g. brisk walking, cycling) 150 min/week. Caution: avoid excessive cardio without RT, which might exacerbate calorie deficit and muscle burn. Balance is key.
Combined exercise (concurrent training)	Aerobic + Resistance yields complementary benefits; maximizes fat loss and muscle retention.	Best evidence for functional improvement in obese older adults. Both modalities together counter sarcopenic obesity.	Can alternate days or combine in sessions. Important to allow adequate recovery (do not overtrain in deficit).
Timing of protein and exercise	Aligns nutrient availability with muscle building windows; exercise in fed state for performance.	Theory-based: post-exercise protein is known to enhance muscle repair; no direct GLP-1 studies but general sports science applies.	Advise protein intake (20–30 g) within 1 h post-RT. If GLP-1 causes nausea in morning, perhaps exercise later when can eat after. Individualize.
Physical therapy/functional training	Preserves neuromuscular function; targets balance, mobility in addition to muscle mass.	Shown to reduce fall risk and improve ADLs in weight loss programmes for older adults. Strength + balance training is superior to strength alone for preventing functional decline.	For older or frail patients, include PT-led exercises. Monitor functional metrics (gait speed, chair rise time).
Ensure micronutrients (vitamin D, etc.)	Supports muscle function and metabolism. Vitamin D deficiency, for example, can impair muscle strength.	If deficient, replacing vitamin D improves muscle function in elderly; consider omega-3 (anti-inflammatory, may aid muscle in cachexia).	Check 25(OH)D levels; supplement to >30 ng/mL if low. General multivitamin to cover bases given reduced food intake on GLP-1 RAs.

25(OH)D, 25-hydroxyvitamin D; ADL, activities of daily living; GLP-1, glucagon-like peptide-1; RA, glucagon-like peptide-1 receptor agonist; HMB, β -hydroxy β -methylbutyrate; mTOR, mechanistic target of rapamycin; PT, physical therapy/physical therapist; RT, resistance training.

RAD 140 (Testolone, RAD-140, Vosilasarm) in rodents and non-human primates increased lean mass and bone density and spares off-target androgenic effects showing tissue selectivity without stimulating prostate enlargement, whereas, in cynomolgus monkeys, a 28-day dosing regimen (0.1–1 mg/kg/day) produced over 10% weight gain, primarily lean mass, with no increase in fat mass, demonstrating anabolic potency and metabolic selectivity.¹⁴⁶ Early human data suggest it promotes fat loss and preserves lean mass.^{149,150}

GSK2881078 increases lean mass in catabolic states. In a phase 1b trial, GSK2881078 (0.75–4 mg/day) increased lean mass without androgenic effects, showing greater sensitivity in women, and mild, transient changes in liver enzymes and high-density lipoprotein cholesterol.¹⁴⁷ A phase 2a trial in chronic obstructive pulmonary disease patients with muscle weakness and active weight loss found increase in lean body mass and muscle strength in both sexes.¹⁴⁸

Anabolic/catabolic transforming agent

S-pindolol benzoate is an orally bioavailable anti-catabolic and pro-anabolic transforming agent (ACTA) that combines partial β_2 -adrenergic agonism, non-selective β -adrenergic antagonism, and 5-HT_{1A} receptor agonism to preserve muscle mass and improve metabolic resilience.^{151,152} In obesity, S-pindolol is being evaluated in the phase 2a PROACT (Preserving, Restoring, and Optimising lean mass and muscle with ACTAs) clinical programme, comprising two randomized, double-blind trials (PROACT-1 and PROACT-2), to assess its capacity to protect muscle during and after GLP-1 RA therapy.¹⁵³ In PROACT-1, patients with obesity will receive S-pindolol benzoate or placebo concomitantly with blinded semaglutide to evaluate changes in lean and fat mass during active weight loss. In PROACT-2, S-pindolol will be initiated after cessation of semaglutide therapy to assess its ability to prevent disproportionate fat regain and support lean mass restoration. Notably, the PROACT programme employs a rigorous double-dummy, fully blinded design, including blinded GLP-1 RA administration, a feature rarely implemented in GLP-1-based studies. Secondary endpoints include detailed assessment of functional performance, cardiac structure and function, and bone composition, enabling evaluation of how S-pindolol may modify the cardiovascular effects of GLP-1 RA-induced weight loss.

Adjunctive strategies to prevent muscle loss

Adjunctive interventions are important for preserving skeletal muscle during incretin-based weight loss therapy³⁶ (Table 4). Resistance training, which stimulates muscle protein synthesis via mTORC1 activation, remains the first-line and most clinically accessible strategy to attenuate muscle loss when combined with hypocaloric diets.¹⁵⁴ Beyond preserving lean mass, resistance exercise improves muscle strength, neuromuscular function, and insulin sensitivity, and should be routinely prescribed alongside anti-obesity pharmacotherapy to optimize functional and cardio-metabolic outcomes. Guidelines recommend resistance training

2–3 times per week for muscle preservation during weight loss.¹⁵⁵ Adequate dietary protein intake in the range of 1.2–1.5 g/kg/day support muscle protein synthesis and mitigates catabolic losses during weight reduction and supplementation with whey or casein recommended for individuals with diminished appetite.¹⁵⁶ These recommendations are extrapolated from general weight-loss literature; GLP-1-specific guideline or trial data are currently lacking for diet and exercise, and both exercise and diet strategies should be individualized, particularly in patients with advanced heart failure and chronic kidney disease.^{157,158} Leucine and its metabolite β -hydroxy β -methylbutyrate can enhance anabolic signalling and reduce muscle breakdown.¹⁵⁹ Creatine monohydrate (3–5 g/day) may help preserve lean mass during weight loss.¹⁶⁰ Aerobic exercise enhances fat oxidation and synergizes with resistance training, and show superior improvements in body composition and functional performance.¹⁶¹ Ensuring adequate intake of micronutrients, vitamins, and omega-3 fatty acids may supports muscle metabolism.¹⁶² In frail or immobile patients, passive modalities like neuromuscular electrical stimulation may offer alternative means of muscle activation.¹⁶³ Lifestyle, nutritional and assessment strategies do more than preserve muscle mass. They enhance weight-loss benefits by lowering blood pressure, improving lipids, enhancing insulin sensitivity, and reducing cardiovascular risk.

Future direction

The evolving focus on body composition in obesity therapy for cardiovascular prevention has highlighted therapeutic challenges and gaps in our understanding of medication effects. There is a need to redefine efficacy endpoints,^{164,165} as conventional metrics focus on weight reduction and neglect lean and muscle mass and functional performance.¹⁶⁶ Greater muscle volume does not necessarily equate to improved strength or physical function, the outcomes of greatest clinical relevance. Future evaluations should prioritize assessment of muscle volume, quality, and function to strengthen the evidence linking body composition with cardiovascular outcomes. Validated methods such as DEXA for appendicular lean mass, CT/MRI for visceral adipose tissue and muscle quality, and MRI-proton density fat fraction for liver fat, help distinguish lean mass from skeletal muscle and to harmonize evidence linking body-composition changes with cardiovascular outcomes. Costs, access barriers, and weight regain off therapy highlight the need for chronic use, careful titration, and equitable access policies. Performance metrics like grip strength assessment offers guidance for targeted interventions.¹⁶⁷ Readily accessible means of monitoring muscle mass and function should be integrated into clinical care. Educating patients on the value of high-quality weight loss losing fat and preserving muscle can improve adherence and outcomes.³⁶ Long-term trials should evaluate combination regimens with lean mass preserving strategies to enhance weight maintenance, metabolic health, functional independence, and maximize cardiovascular outcomes. Next-generation trials of incretin therapies should prespecify body-composition and fitness endpoints.

Conclusions

Newer obesity drugs have redefined obesity treatment and have demonstrated substantial cardiometabolic and cardiovascular

benefits. However, this comes with lean mass loss, which may compromise strength, function, and long-term metabolic resilience. The quality of weight loss should therefore prioritize fat reduction while preserving muscle and physical function, rather than muscle quantity alone. Emerging pharmacologic strategies targeting pathways of muscle preservation or anabolism represent promising avenues to complement incretin-based therapies, warranting well-designed future trials to define their clinical role in optimizing body composition with pharmacological weight loss. Lifestyle interventions, resistance training and adequate protein intake remain the cornerstone for muscle preservation. Future obesity care must integrate strategies focusing on cardiovascular outcomes, functionality, and quality of life and shift goals from pounds lost to health gained. A body composition focused approach that protects lean mass and reduces adiposity promises to offer the greatest benefit.

Acknowledgements

Editorial Support was provided by MediCog LLC.

Supplementary data

Supplementary data are not available at [European Heart Journal](https://academic.oup.com/eurheartj/article/47/21/2584/8566002) online.

Declarations

Disclosure of Interest

MSK has received fees from Bayer, Boehringer Ingelheim and Novartis. **MHD** has no conflicts of interest to declare in relation to this work. **YH** has received research grants or consultant and speaker honoraria from Amgen, Applied Therapeutics, AstraZeneca, Bayer, Boehringer Ingelheim, Corcept Therapeutics, Esperion, Ionis, Lilly, Mankind, Merck, Merck-Pfizer, Novartis, Novo Nordisk, and Sanofi. **SDA** reported receiving grants from Abbott Laboratories; receiving personal fees from Actimed Therapeutics, Alleviant, AstraZeneca, Bayer, Berlin Heals, BioVentric, Boehringer Ingelheim, Brahms, Cardiac Dimensions, Cardior Pharmaceuticals GmbH, Cordio, CSL Vifor, CVRx, Cytokinetics, Edwards Lifesciences, Farraday Pharmaceuticals, GSK, HeartKinetics, Impulse Dynamics, Lilly, Mankind Pharma, Medtronic, Novartis, Novo Nordisk, Occlutech, Pfizer, Regeneron, Relaxera, Repairon GmbH, SCIRENT Clinical Research and Science, Sensible Medical, Servier, Vectorious Medical Technologies, Vivus, and V-Wave; and being a named co-inventor of 2 patent applications regarding midregional proatrial natriuretic peptide, but does not benefit personally from the related issued patents. **AJSC** received grants and personal fees from Vifor International and personal fees from AstraZeneca, Bayer, Boehringer Ingelheim, Menarini, Novartis, Nutricia, Servier, Vifor, Abbott, Actimed, Arena, Cardiac Dimensions, Corvia, CVRx, Enopace, ESN Cleer, Faraday, Gore, Impulse Dynamics, and Respicardia. **JBG** reports institutional grant funding from Merck, Roche, Boehringer Ingelheim, Eli Lilly, Bluedrop and consulting fees from AstraZeneca, Novo Nordisk, Pfizer, Bayer, Anji, Boehringer Ingelheim, Valo, Eli Lilly, Vertex and Mineralys. **JB** has received payment or honoraria for lectures, presentations, Speakers

Bureau, manuscript writing, or educational events from Abbott, Adrenomed, Amgen, Applied Therapeutics, Array, AstraZeneca, Bayer, BerlinCures, Cardior, CVRx, Foundry, G3 Pharma, Imbria, Impulse Dynamics, Innolife, Janssen, LivaNova, Luitpold, Medtronic, Merck, Novartis, Novo Nordisk, Relypsa, Roche, Sanofi, Sequana Medical, Occlutech, and Vifor, and is a Trial Executive Committee member of Boehringer Ingelheim and Eli Lilly and Company Diabetes Alliance.

Data Availability

No data were generated or analysed for or in support of this paper.

Funding

Nothing to declare.

References

- Bray GA, Kim KK, Wilding JPH; World Obesity Federation. Obesity: a chronic relapsing progressive disease process. A position statement of the World Obesity Federation. *Obes Rev* 2017;**18**:715–23. <https://doi.org/10.1111/obr.12551>
- World Health Organization. *Obesity and Overweight* [Internet]. Geneva: World Health Organization, 2025. <https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight> (1 September 2025, date last accessed).
- GBD 2021 Risk Factors Collaborators. Global burden and strength of evidence for 88 risk factors in 204 countries and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet*. 2024;**403**:2162–203. [https://doi.org/10.1016/S0140-6736\(24\)00933-4](https://doi.org/10.1016/S0140-6736(24)00933-4)
- Okunogbe A, Nugent R, Spencer G, Powis J, Ralston J, Wilding J. Economic impacts of overweight and obesity: current and future estimates for 161 countries. *BMJ Glob Health* 2022;**7**:e009773. <https://doi.org/10.1136/bmjgh-2022-009773>
- World Obesity Federation. *Economic Impact of Overweight and Obesity to Surpass \$4 Trillion by 2035* [Internet]. World Obesity Federation, 2023. <https://www.worldobesity.org/news/economic-impact-of-overweight-and-obesity-to-surpass-4-trillion-by-2035> (1 September 2025, date last accessed).
- Blüher M. Obesity: global epidemiology and pathogenesis. *Nat Rev Endocrinol* 2019;**15**:288–98. <https://doi.org/10.1038/s41574-019-0176-8>
- Koskinas KC, Van Craenenbroeck EM, Antoniadou C, Blüher M, Gorter TM, Hanssen H, et al. Obesity and cardiovascular disease: an ESC clinical consensus statement. *Eur Heart J* 2024;**45**:4063–98. <https://doi.org/10.1093/eurheartj/ehae508>
- World Obesity Federation. *Economic Impact of Overweight and Obesity to Surpass \$4 Trillion by 2035* [Internet]. World Obesity Federation, 2023. <https://www.worldobesity.org/news/economic-impact-of-overweight-and-obesity-to-surpass-4-trillion-by-2035> (8 February 2026, date last accessed).
- Wadden TA, Tronieri JS, Butryn ML. Lifestyle modification approaches for the treatment of obesity in adults. *Am Psychol* 2020;**75**:235–51. <https://doi.org/10.1037/amp0000517>
- Kushner RF. Weight loss strategies for treatment of obesity: lifestyle management and pharmacotherapy. *Prog Cardiovasc Dis* 2018;**61**:246–52. <https://doi.org/10.1016/j.pcad.2018.06.001>
- Qi QYD, Cox A, McNeil S, Sumithran P. Obesity medications: a narrative review of current and emerging agents. *Osteoarthritis Cartil Open* 2024;**6**:100472. <https://doi.org/10.1016/j.ocarto.2024.100472>
- Forst T, De Block C, Del Prato S, Frias J, Lautenbach A, Ludvik B, et al. Novel pharmacotherapies for weight loss: understanding the role of incretins to enable weight loss and improved health outcomes. *Diabetes Obes Metab* 2025;**27**:48–65. <https://doi.org/10.1111/dom.16247>
- Enright C, Thomas E, Saxon DR. An updated approach to antiobesity pharmacotherapy: moving beyond the 5% weight loss goal. *J Endocr Soc* 2023;**7**:bvac195. <https://doi.org/10.1210/jendso/bvac195>
- Song JE, Ko HJ, Kim AS. Comparison of the efficacy of anti-obesity medications in real-world practice. *Drug Des Devel Ther* 2024;**18**:845–58. <https://doi.org/10.2147/DDDT.S445415>

15. Idrees Z, Cancarevic I, Huang L. FDA-approved pharmacotherapy for weight loss over the last decade. *Cureus* 2022;**14**:e29262. <https://doi.org/10.7759/cureus.29262>
16. Patel DK, Stanford FC. Safety and tolerability of new-generation anti-obesity medications: a narrative review. *Postgrad Med* 2018;**130**:173–82. <https://doi.org/10.1080/00325481.2018.1435129>
17. Pi-Sunyer X, Astrup A, Fujioka K, Greenway F, Halpern A, Krempf M, et al. A randomized, controlled trial of 3.0 mg of liraglutide in weight management. *N Engl J Med* 2015;**373**:11–22. <https://doi.org/10.1056/NEJMoa1411892>
18. Bergmann NC, Davies MJ, Lingway I, Knop FK. Semaglutide for the treatment of overweight and obesity: a review. *Diabetes Obes Metab* 2023;**25**:18–35. <https://doi.org/10.1111/dom.14863>
19. Horn DB, Kahan S, Batterham RL, Cao D, Lee CJ, Murphy M, et al. Time to weight plateau with tirzepatide treatment in the SURMOUNT-1 and SURMOUNT-4 clinical trials. *Clin Obes* 2025;**15**:e12734. <https://doi.org/10.1111/cob.12734>
20. Latest Data From SURPASS Trials Demonstrate Tirzepatide Provided Meaningful Blood Sugar Reductions [press release]. Diabetes.org Newsroom. American Diabetes Association, 2021. <https://diabetes.org/newsroom/latest-data-from-SURPASS-trials-demonstrate-tirzepatide-provided-blood-sugar-reductions-weight-loss> (1 September 2025, date last accessed).
21. Viking Therapeutics Announces Initiation of Phase 2 VENTURE-Oral Dosing Trial of VK2735 Tablet Formulation in Patients With Obesity [Internet]. Viking Therapeutics, 2025. <https://ir.vikingtherapeutics.com/2025-01-08-Viking-Therapeutics-Announces-Initiation-of-Phase-2-VENTURE-Oral-Dosing-Trial-of-VK2735-Tablet-Formulation-in-Patients-with-Obesity> (1 September 2025, date last accessed).
22. Psaltis JP, Marathe JA, Nguyen MT, Le R, Bursill CA, Marathe CS, et al. Incretin-based therapies for the management of cardiometabolic disease in the clinic: past, present, and future. *Med Res Rev* 2025;**45**:29–65. <https://doi.org/10.1002/med.22070>
23. Christensen S, Robinson K, Thomas S, Williams DR. Dietary intake by patients taking GLP-1 and dual GIP/GLP-1 receptor agonists: a narrative review and discussion of research needs. *Obes Pillars* 2024;**11**:100121. <https://doi.org/10.1016/j.obpill.2024.100121>
24. Argyrakopoulou G, Gitsi E, Konstantinidou SK, Kokkinos A. The effect of obesity pharmacotherapy on body composition, including muscle mass. *Int J Obes (Lond)* 2025;**49**:381–7. <https://doi.org/10.1038/s41366-024-01533-3>
25. Tchang BG, Aras M, Kumar RB, Aronne LJ. Pharmacologic treatment of overweight and obesity in adults [Updated 20 August 2024]. In: Feingold KR, Ahmed SF, Anawalt B, et al. (eds.), *Endotext* [Internet]. South Dartmouth (MA): MDTtext.com, Inc.; 2000. <https://www.ncbi.nlm.nih.gov/books/NBK279038/> (1 September 2025, date last accessed).
26. Horváth L, Mráz M, Jude EB, Haluzik M. Pharmacotherapy as an augmentation to bariatric surgery for obesity. *Drugs* 2024;**84**:933–52. <https://doi.org/10.1007/s40265-024-02029-0>
27. Moiz A, Filion KB, Tsoukas MA, Yu OH, Peters TM, Eisenberg MJ. Mechanisms of GLP-1 receptor agonist-induced weight loss: a review of central and peripheral pathways in appetite and energy regulation. *Am J Med* 2025;**138**:934–40. <https://doi.org/10.1016/j.amjmed.2025.01.021>
28. Hope DCD, Tan TMM. Skeletal muscle loss and sarcopenia in obesity pharmacotherapy. *Nat Rev Endocrinol* 2024;**20**:695–6. <https://doi.org/10.1038/s41574-024-01041-4>
29. Stefanakis K, Kokkorakis M, Mantzoros CS. The impact of weight loss on fat-free mass, muscle, bone and hematopoiesis health: implications for emerging pharmacotherapies aiming at fat reduction and lean mass preservation. *Metabolism* 2024;**161**:156057. <https://doi.org/10.1016/j.metabol.2024.156057>
30. Kim CH, Wheatley CM, Behnia M, Johnson BD. The effect of aging on relationships between lean body mass and VO₂max in Rowers. *PLoS One* 2016;**11**:e0160275. <https://doi.org/10.1371/journal.pone.0160275>
31. Sylow L, Tokarz VL, Richter EA, Klip A. The many actions of insulin in skeletal muscle, the paramount tissue determining glycemia. *Cell Metab* 2021;**33**:758–80. <https://doi.org/10.1016/j.cmet.2021.03.020>
32. Prado CM, Phillips SM, Gonzalez MC, Heymsfield SB. Muscle matters: the effects of medically induced weight loss on skeletal muscle. *Lancet Diabetes Endocrinol* 2024;**12**:785–7. [https://doi.org/10.1016/S2213-8587\(24\)00272-9](https://doi.org/10.1016/S2213-8587(24)00272-9)
33. Cava E, Yeat NC, Mittendorfer B. Preserving healthy muscle during weight loss. *Adv Nutr* 2017;**8**:511–9. <https://doi.org/10.3945/an.116.014506>
34. McCarthy D, Berg A. Weight loss strategies and the risk of skeletal muscle mass loss. *Nutrients* 2021;**13**:2473. <https://doi.org/10.3390/nu13072473>
35. Olson B, Marks DL, Grossberg AJ. Diverging metabolic programmes and behaviours during states of starvation, protein malnutrition, and cachexia. *J Cachexia Sarcopenia Muscle* 2020;**11**:1429–46. <https://doi.org/10.1002/jcsm.12630>
36. Mechanick JL, Butsch WS, Christensen SM, Hamdy O, Li Z, Prado CM, et al. Strategies for minimizing muscle loss during use of incretin-mimetic drugs for treatment of obesity. *Obes Rev* 2025;**26**:e13841. <https://doi.org/10.1111/obr.13841>
37. Margolis LM, Rivas DA, Berrone M, Ezzyat Y, Young AJ, McClung JP, et al. Prolonged calorie restriction downregulates skeletal muscle mTORC1 signaling independent of dietary protein intake and associated microRNA expression. *Front Physiol* 2016;**7**:445. <https://doi.org/10.3389/fphys.2016.00445>
38. Mather KJ, Mari A, Heise T, DeVries JH, Hua M, Urva S, et al. Effects of tirzepatide vs semaglutide on β -cell function, insulin sensitivity, and glucose control during a meal test. *J Clin Endocrinol Metab* 2024;**109**:3046–54. <https://doi.org/10.1210/clinem.dgae319>
39. Frost RA, Lang CH. Multifaceted role of insulin-like growth factors and mammalian target of rapamycin in skeletal muscle. *Endocrinol Metab Clin North Am* 2012;**41**:297–322, vi. <https://doi.org/10.1016/j.ecl.2012.04.012>
40. Fuentes EN, Björnsson BT, Valdés JA, Einarsson IE, Lorca B, Alvarez M, et al. IGF-1/PI3K/Akt and IGF-1/MAPK/ERK pathways in vivo in skeletal muscle are regulated by nutrition and contribute to somatic growth in the fine flounder. *Am J Physiol Regul Integr Comp Physiol* 2011;**300**:R1532–R42. <https://doi.org/10.1152/ajpregu.00535.2010>
41. Barclay RD, Burd NA, Tyler C, Tillin NA, Mackenzie RW. The role of the IGF-1 signaling cascade in muscle protein synthesis and anabolic resistance in aging skeletal muscle. *Front Nutr* 2019;**6**:146. <https://doi.org/10.3389/fnut.2019.00146>
42. Chen K, Gao P, Li Z, Dai A, Yang M, Chen S, et al. Forkhead Box O signaling pathway in skeletal muscle atrophy. *Am J Pathol* 2022;**192**:1648–57. <https://doi.org/10.1016/j.ajpath.2022.09.003>
43. Stitt TN, Drujan D, Clarke BA, Panaro F, Timofeyeva Y, Kline WO, et al. The IGF-1/PI3K/Akt pathway prevents expression of muscle atrophy-induced ubiquitin ligases by inhibiting FOXO transcription factors. *Mol Cell* 2004;**14**:395–403. [https://doi.org/10.1016/s1097-2765\(04\)00211-4](https://doi.org/10.1016/s1097-2765(04)00211-4)
44. Elkina Y, von Haehling S, Anker SD, Springer J. The role of myostatin in muscle wasting: an overview. *J Cachexia Sarcopenia Muscle* 2011;**2**:143–51. <https://doi.org/10.1007/s13539-011-0035-5>
45. Linge J, Birkenfeld AL, Neeland IJ. Muscle mass and glucagon-like peptide-1 receptor agonists: adaptive or maladaptive response to weight loss? *Circulation* 2024;**150**:1288–98. <https://doi.org/10.1161/CIRCULATIONAHA.124.067676>
46. Liu Z, Weeldreyer NR, Angadi SS. Incretin receptor agonism, fat-free mass, and cardiorespiratory fitness: a narrative review. *J Clin Endocrinol Metab* 2025;**110**:2709–17. <https://doi.org/10.1210/clinem.dgaf335>
47. Ussher JR, Drucker DJ. Cardiovascular biology of the incretin system. *Endocr Rev* 2012;**33**:187–215. <https://doi.org/10.1210/er.2011-1052>
48. Nuijten MAH, Eijssvogels TMH, Montpellier VM, Janssen IMC, Hazebroek EJ, Hopman MTE. The magnitude and progress of lean body mass, fat-free mass, and skeletal muscle mass loss following bariatric surgery: a systematic review and meta-analysis. *Obes Rev* 2022;**23**:e13370. <https://doi.org/10.1111/obr.13370>
49. Christensen S, Nelson C. Chronicity of obesity and the importance of early treatment to reduce cardiometabolic risk and improve body composition. *Obes Pillars* 2025;**15**:100175. <https://doi.org/10.1016/j.obpill.2025.100175>
50. Chait A, den Hartigh LJ. Adipose tissue distribution, inflammation and its metabolic consequences, including diabetes and cardiovascular disease. *Front Cardiovasc Med* 2020;**7**:22. <https://doi.org/10.3389/fcvm.2020.00022>
51. Santini S, Vionnet N, Pasquier J, Gonzalez-Rodriguez E, Fraga M, Pitteloud N, et al. Marked weight loss on liraglutide 3.0 mg: real-life experience of a Swiss cohort with obesity. *Obesity (Silver Spring)*. 2023;**31**:74–82. <https://doi.org/10.1002/oby.23596>
52. Silver HJ, Olson D, Mayfield D, Wright P, Nian H, Mashayekhi M, et al. Effect of the glucagon-like peptide-1 receptor agonist liraglutide, compared to caloric restriction, on appetite, dietary intake, body fat distribution and cardio-metabolic biomarkers: a randomized trial in adults with obesity and prediabetes. *Diabetes Obes Metab* 2023;**25**:2340–50. <https://doi.org/10.1111/dom.15113>
53. Yu DN, Wang LJ, Cheng B, Li M, Pan Q, Guo LX. The effects of liraglutide on body composition and muscle strength in adult obese patients with type 2 diabetes mellitus. *Zhonghua Nei Ke Za Zhi* 2021;**60**:982–6. <https://doi.org/10.3760/cma.j.cn112138-20210205-00105>
54. Ghanim H, Batra M, Green K, Abuayseh S, Hejna J, Makdissi A, et al. Liraglutide treatment in overweight and obese patients with type 1 diabetes: a 26-week randomized controlled trial; mechanisms of weight loss. *Diabetes Obes Metab* 2020;**22**:1742–52. <https://doi.org/10.1111/dom.14090>
55. Wilding JPH, Batterham RL, Calanna S, Van Gaal LF, McGowan BM, Rosenstock J, et al. Impact of semaglutide on body composition in adults

- with overweight or obesity: exploratory analysis of the STEP 1 study. *J Endocr Soc* 2021;**5**:A16–7. <https://doi.org/10.1210/jendso/bvab048.030>
56. McCrimmon RJ, Catarig AM, Frias JP, Lausvig NL, le Roux CW, Thielke D, et al. Effects of once-weekly semaglutide vs once-daily canagliflozin on body composition in type 2 diabetes: a substudy of the SUSTAIN 8 randomised controlled clinical trial. *Diabetologia* 2020;**63**:473–85. <https://doi.org/10.1007/s00125-019-05065-8>
 57. Rodríguez Jiménez B, Rodríguez de Vera Gómez P, Belmonte Lomas S, Mesa Díaz AM, Caballero Mateos I, Galán I, et al. Transforming body composition with semaglutide in adults with obesity and type 2 diabetes mellitus. *Front Endocrinol (Lausanne)* 2024;**15**:1386542. <https://doi.org/10.3389/fendo.2024.1386542>
 58. Uchiyama S, Sada Y, Mihara S, Sasaki Y, Sone M, Tanaka Y. Oral semaglutide induces loss of body fat mass without affecting muscle mass in patients with type 2 diabetes. *J Clin Med Res* 2023;**15**:377–83. <https://doi.org/10.14740/jocmr4987>
 59. Volpe S, Lisco G, Fanelli M, Racaniello D, Colaianni V, Lavarra V, et al. Oral semaglutide improves body composition and preserves lean mass in patients with type 2 diabetes: a 26-week prospective real-life study. *Front Endocrinol* 2023;**14**:1240263. <https://doi.org/10.3389/fendo.2023.1240263>
 60. Chun E, Siojo A, Rivera D, Reyna K, Legere H, Joseph R, et al. Weight loss and body composition after compounded semaglutide treatment in a real world setting. *Diabetes Obes Metab* 2025;**27**:1536–43. <https://doi.org/10.1111/dom.16162>
 61. Garvey WT, Blüher M, Osorto Contreras CK, Davies MJ, Winning Lehmann E, Pietiläinen KH, et al. Coadministered cagrilintide and semaglutide in adults with overweight or obesity. *N Engl J Med* 2025;**393**:635–47. <https://doi.org/10.1056/NEJMoa2502081>
 62. Look M, Dunn JP, Kushner RF, Cao D, Harris C, Gible TH, et al. Body composition changes during weight reduction with tirzepatide in the SURMOUNT-1 study of adults with obesity or overweight. *Diabetes Obes Metab* 2025;**27**:2720–9. <https://doi.org/10.1111/dom.16275>
 63. Zinn J, Poretsky L. Skeletal muscle mass and body weight fall proportionally with use of dual glucagon-like peptide 1/glucose-dependent insulinotropic polypeptide receptor agonist tirzepatide: case report and review of literature. *AACE Clin Case Rep* 2025;**11**:98–101. <https://doi.org/10.1016/j.aace.2024.12.001>
 64. Adamidis N, Desalermos A, Papadopoulou N, Dalianis A, Bantis A, Tsiakos K, et al. Short-term effects of tirzepatide in obese adults: a real-world prospective study. *Cureus* 2025;**17**:e85970. <https://doi.org/10.7759/cureus.85970>
 65. Sattar N, Neeland IJ, Dahlqvist Leinhard O, Fernández Landó L, Bray R, Linge J, et al. Tirzepatide and muscle composition changes in people with type 2 diabetes (SURPASS-3 MRI): a post-hoc analysis of a randomised, open-label, parallel-group, phase 3 trial. *Lancet Diabetes Endocrinol* 2025;**13**:482–93. [https://doi.org/10.1016/S2213-8587\(25\)00027-0](https://doi.org/10.1016/S2213-8587(25)00027-0)
 66. Merz KE, Thurmond DC. Role of skeletal muscle in insulin resistance and glucose uptake. *Compr Physiol* 2020;**10**:785–809. <https://doi.org/10.1002/cphy.c190029>
 67. Greenway FL. Semaglutide and tirzepatide to treat obesity. *N Engl J Med* 2025;**393**:84–5. <https://doi.org/10.1056/NEJMe2507452>
 68. Karakasis P, Patoulas D, Fragakis N, Mantzoros CS. Effect of glucagon-like peptide-1 receptor agonists and co-agonists on body composition: systematic review and network meta-analysis. *Metabolism* 2025;**164**:156113. <https://doi.org/10.1016/j.metabol.2024>
 69. Severinsen MCK, Pedersen BK. Muscle-organ crosstalk: the emerging roles of myokines. *Endocr Rev* 2020;**41**:594–609. <https://doi.org/10.1210/endo/bnaa024>
 70. Shero JA, Lindholm ME, Sandri M, Stanford KI. Skeletal muscle as a mediator of interorgan crosstalk during exercise: implications for aging and obesity. *Circ Res* 2025;**136**:1407–32. <https://doi.org/10.1161/CIRCRESAHA.124.325614>
 71. Muoio Deborah M, Neuffer PD. Lipid-induced mitochondrial stress and insulin action in muscle. *Cell Metab* 2012;**15**:595–605. <https://doi.org/10.1016/j.cmet.2012.04.010>
 72. Usman MS, Davies M, Hall ME, Verma S, Anker SD, Rosenstock J, et al. The cardiovascular effects of novel weight loss therapies. *Eur Heart J* 2023;**44**:5036–48. <https://doi.org/10.1093/eurheartj/ehad664>
 73. Marso SP, Daniels GH, Brown-Frandsen K, Kristensen P, Mann JFE, Nauck MA, et al. Liraglutide and cardiovascular outcomes in type 2 diabetes. *N Engl J Med* 2016;**375**:311–22. <https://doi.org/10.1056/NEJMoa1603827>
 74. Lincoff AM, Brown-Frandsen K, Colhoun HM, Deanfield J, Emerson SS, Esbjerg S, et al. Semaglutide and cardiovascular outcomes in obesity without diabetes. *N Engl J Med* 2023;**389**:2221–32. <https://doi.org/10.1056/NEJMoa2307563>
 75. Marx N, Federici M, Schütt K, Müller-Wieland D, Ajjan RA, Antunes MJ, et al. 2023 ESC guidelines for the management of cardiovascular disease in patients with diabetes. *Eur Heart J* 2023;**44**:4043–140. <https://doi.org/10.1093/eurheartj/ehad192>
 76. Verma S, McGuire DK, Bain SC, Bhatt DL, Leiter LA, Mazer CD, et al. Effects of glucagon-like peptide-1 receptor agonists liraglutide and semaglutide on cardiovascular and renal outcomes across body mass index categories in type 2 diabetes: results of the LEADER and SUSTAIN 6 trials. *Diabetes Obes Metab* 2020;**22**:2487–92. <https://doi.org/10.1111/dom.14160>
 77. Gerstein HC, Colhoun HM, Dagenais GR, Diaz R, Lakshmanan M, Pais P, et al. Dulaglutide and cardiovascular outcomes in type 2 diabetes (REWIND): a double-blind, randomised placebo-controlled trial. *Lancet* 2019;**394**:121–30. [https://doi.org/10.1016/S0140-6736\(19\)31149-3](https://doi.org/10.1016/S0140-6736(19)31149-3)
 78. Husain M, Bain SC, Jeppesen OK, Lingvay I, Sørrig R, Treppendahl MB, et al. Semaglutide (SUSTAIN and PIONEER) reduces cardiovascular events in type 2 diabetes across varying cardiovascular risk. *Diabetes Obes Metab* 2020;**22**:442–51. <https://doi.org/10.1111/dom.13955>
 79. Kosiborod MN, Petrie MC, Borlaug BA, Butler J, Davies MJ, Hovingh GK, et al. Semaglutide in patients with obesity-related heart failure and type 2 diabetes. *N Engl J Med* 2024;**390**:1394–407. <https://doi.org/10.1056/NEJMoa2313917>
 80. Arty F, Shreya D, Chaudhry A, Sohini S. Semaglutide in heart failure with preserved ejection fraction: emerging evidence and clinical implications. *Cureus* 2025;**17**:e87605. <https://doi.org/10.7759/cureus.87605>
 81. Rehman A, Saidullah S, Asad M, Gondal UR, Ashraf A, Khan MF, et al. Efficacy and safety of semaglutide in patients with heart failure with preserved ejection fraction and obesity. *Clin Cardiol* 2024;**47**:e24283. <https://doi.org/10.1002/clc.24283>
 82. Kosiborod MN, Abildstrøm SZ, Borlaug BA, Butler J, Rasmussen S, Davies M, et al. Semaglutide in patients with heart failure with preserved ejection fraction and obesity. *N Engl J Med* 2023;**389**:1069–84. <https://doi.org/10.1056/NEJMoa2306963>
 83. Packer M, Zile MR, Kramer CM, Baum SJ, Litwin SE, Menon V, et al. Tirzepatide for heart failure with preserved ejection fraction and obesity. *N Engl J Med* 2025;**392**:427–37. <https://doi.org/10.1056/NEJMoa2410027>
 84. Nicholls SJ, Pavo I, Bhatt DL, Buse JB, Prato SD, Kahn SE, et al. Cardiovascular outcomes with tirzepatide versus dulaglutide in type 2 diabetes. *N Engl J Med* 2025;**393**:2409–20. <https://doi.org/10.1056/NEJMoa2505928>
 85. Neeland IJ, Linge J, Birkenfeld AL. Changes in lean body mass with glucagon-like peptide-1-based therapies and mitigation strategies. *Diabetes Obes Metab* 2024;**26**:16–27. <https://doi.org/10.1111/dom.15728>
 86. Kreitzman SN, Coxon AY, Szaz KF. Glycogen storage: illusions of easy weight loss, excessive weight regain, and distortions in estimates of body composition. *Am J Clin Nutr* 1992;**56**:292S–3S. <https://doi.org/10.1093/ajcn/56.1.292S>
 87. Toomey CM, McCormack WG, Jakeman P. The effect of hydration status on the measurement of lean tissue mass by dual-energy X-ray absorptiometry. *Eur J Appl Physiol* 2017;**117**:567–74. <https://doi.org/10.1007/s00421-017-3552-x>
 88. Bastly N, Thanaj M, Whitcher B, Bell JD, Thomas EL. Comparing DXA and MRI body composition measurements in cross-sectional and longitudinal cohorts. *Commun Med* 2026. <https://doi.org/10.1038/s43856-026-01440-w>
 89. Tavoian D, Ampomah K, Amano S, Law TD, Clark BC. Changes in DXA-derived lean mass and MRI-derived cross-sectional area of the thigh are modestly associated. *Sci Rep* 2019;**9**:10028. <https://doi.org/10.1038/s41598-019-46428-w>
 90. Baglietto N, Vaquero-Cristóbal R, Albaladejo-Saura M, Mecherques-Carini M, Esparza-Ros F. Assessing skeletal muscle mass and lean body mass: an analysis of the agreement among dual X-ray absorptiometry, anthropometry, and bioelectrical impedance. *Front Nutr* 2024;**11**:1445892. <https://doi.org/10.3389/fnut.2024.1445892>
 91. Brennan AM, Standley RA, Anthony SJ, Grench KE, Helbling NL, DeLany JP, et al. Weight loss and exercise differentially affect insulin sensitivity, body composition, cardiorespiratory fitness, and muscle strength in older adults with obesity: a randomized controlled trial. *J Gerontol A Biol Sci Med Sci* 2022;**77**:1088–97. <https://doi.org/10.1093/gerona/glab240>
 92. Soto ME, Pérez-Torres I, Rubio-Ruiz ME, Cano-Martínez A, Manzano-Pech L, Guarner-Lans V. Frailty and the interactions between skeletal muscle, bone, and adipose tissue-impact on cardiovascular disease and possible therapeutic measures. *Int J Mol Sci* 2023;**24**:4534. <https://doi.org/10.3390/ijms24054534>
 93. Gill LE, Bartels SJ, Batsis JA. Weight management in older adults. *Curr Obes Rep* 2015;**4**:379–88. <https://doi.org/10.1007/s13679-015-0161-z>
 94. Bennett JP, Lim S. The critical role of body composition assessment in advancing research and clinical health risk assessment across the lifespan. *J Obes Metab Syndr* 2025;**34**:120–37. <https://doi.org/10.7570/jomes25010>
 95. Martínez MC, Meli EF, Candia FP, Filippi F, Vilallonga R, Cordero E, et al. The impact of bariatric surgery on the muscle mass in patients with obesity:

- 2-year follow-up. *Obes Surg* 2022;**32**:625–33. <https://doi.org/10.1007/s11695-021-05815-x>
96. Coskun T, Wu Q, Schlot NC, Haupt A, Milicevic Z, Khouli C, et al. Effects of retatrutide on body composition in people with type 2 diabetes: a substudy of a phase 2, double-blind, parallel-group, placebo-controlled, randomised trial. *Lancet Diabetes Endocrinol* 2025;**13**:674–84. [https://doi.org/10.1016/S2213-8587\(25\)00092-0](https://doi.org/10.1016/S2213-8587(25)00092-0)
 97. Meloux A, Rochette L, Maza M, Bichat F, Tribouillard L, Cottin Y, et al. Growth differentiation factor-8 (GDF8)/myostatin is a predictor of troponin I peak and a marker of clinical severity after acute myocardial infarction. *J Clin Med* 2020;**9**:116. <https://doi.org/10.3390/jcm9010116>
 98. Rodriguez J, Vernus B, Chelhi I, Cassar-Malek I, Gabillard JC, Hadj Sassi A, et al. Myostatin and the skeletal muscle atrophy and hypertrophy signaling pathways. *Cell Mol Life Sci* 2014;**71**:4361–71. <https://doi.org/10.1007/s00018-014-1689-x>
 99. Goodman CA, Hornberger TA. New roles for Smad signaling and phosphatidic acid in the regulation of skeletal muscle mass. *F1000Prime Rep* 2014;**6**:20. <https://doi.org/10.12703/P6-20>
 100. Ge X, McFarlane C, Vajjala A, Lokireddy S, Ng ZH, Tan CK, et al. Smad3 signaling is required for satellite cell function and myogenic differentiation of myoblasts. *Cell Res* 2011;**21**:1591–604. <https://doi.org/10.1038/cr.2011.72>
 101. Jin Q, Qiao C, Li J, Xiao B, Li J, Xiao X. A GDF11/myostatin inhibitor, GDF11 propeptide-Fc, increases skeletal muscle mass and improves muscle strength in dystrophic mdx mice. *Skelet Muscle* 2019;**9**:16. <https://doi.org/10.1186/s13395-019-0197-y>
 102. Lach-Trifiliev E, Minetti GC, Sheppard K, Ibeunjo C, Feige JN, Hartmann S, et al. An antibody blocking activin type II receptors induces strong skeletal muscle hypertrophy and protects from atrophy. *Mol Cell Biol* 2014;**34**:606–18. <https://doi.org/10.1128/MCB.01307-13>
 103. Mastaitis JW, Gomez D, Raya JG, Li D, Min S, Stec M, et al. GDF8 and activin A blockade protects against GLP-1-induced muscle loss while enhancing fat loss in obese male mice and non-human primates. *Nat Commun* 2025;**16**:4377. <https://doi.org/10.1038/s41467-025-59485-9>
 104. Heymsfield SB, Coleman LA, Miller R, Rooks DS, Laurent D, Petricoul O, et al. Effect of Bimagrumb vs placebo on body fat mass among adults with type 2 diabetes and obesity: a phase 2 randomized clinical trial. *JAMA Netw Open* 2021;**4**:e2033457. <https://doi.org/10.1001/jamanetworkopen.2020.33457>
 105. Attie KM, Heymsfield SB, Mindeholm L, Spruill S, Pruzanski M, Klickstein L. 370-OR: optimized weight loss with Bimagrumb—reduced fat mass with increased muscle mass by appetite-independent mechanisms. *Diabetes* 2023;**72**:S370. <https://doi.org/10.2337/db23-370-OR>
 106. Versanis Bio begins enrolment in BELIEVE Phase IIb obesity treatment study [Internet]. Clinical Trials Arena, 2023. <https://www.clinicaltrialsarena.com/news/versanis-bio-obesity-trial/> (15 September 2025, date last accessed).
 107. A Randomized, Double-Blind, Placebo-Controlled Multi-Center Study of Intravenous Bimagrumb, Alone or in Addition to Open Label Subcutaneous Semaglutide, to Investigate the Efficacy and Safety in Overweight or Obese Men and Women [Internet]. Updated June 13, 2025. <https://clinicaltrials.gov/study/NCT05616013> (15 September 2025, date last accessed).
 108. A Study to Investigate Weight Management With Bimagrumb (LY3985863) and Tirzepatide (LY3298176), Alone or in Combination, in Adults With Obesity or Overweight [Internet]. Updated April 2026. <https://clinicaltrials.gov/study/NCT06643728> (15 September 2025, date last accessed).
 109. Scholar Rock Reports Positive Phase 2 EMBRAZE Trial Results Demonstrating Statistically Significant Preservation of Lean Mass With Apitegromab During Tirzepatide-Induced Weight Loss [Internet]. Yahoo! Finance; 2025 June 18. <https://finance.yahoo.com/news/scholar-rock-reports-positive-phase-110000998.html> (15 September 2025, date last accessed).
 110. Fulham MA, Chapron C, Danehy FT, Streich FC, Liang B, Boston CJ, et al. 302-OR: the antimyostatin antibody SRK-439 promotes healthy body composition in combination with GLP-1RAs in a mouse model of obesity. *Diabetes* 2024;**73**:1711–P. <https://doi.org/10.2337/db24-302-OR>
 111. Chugai Pharmaceutical Co., Ltd. Chugai Announces 2024 2nd Quarter (Interim) Results [press release]. Tokyo: Chugai Pharmaceutical, 2024. https://www.chugai-pharm.co.jp/english/news/detail/20240725160000_1089.html (15 September 2025, date last accessed).
 112. Servais L, Lair LL, Connolly AM, Byrne BJ, Chen KS, Coric V, et al. Taldefgrobep Alfa and the phase 3 RESILIENT trial in spinal muscular atrophy. *Int J Mol Sci* 2024;**25**:10273. <https://doi.org/10.3390/ijms251910273>
 113. Biohaven Pharmaceuticals, Inc. Biohaven Drug Misses in Phase III for SMA, Will Be Tested in Obesity [Internet]. New York: BioSpace, 2024 [cited 2025 Jun 16]. <https://www.biospace.com/drug-development/biohaven-drug-misses-in-phase-iii-for-sma-will-be-tested-in-obesity> (15 September 2025, date last accessed).
 114. Interim Results from Ongoing Phase 2 COURAGE Trial Confirm Potential to Improve the Quality of Semaglutide (GLP-1 Receptor Agonist)-Induced Weight Loss by Preserving Lean Mass [press release]. Tarrytown (NY): Regeneron Pharmaceuticals, 2025 [cited 2025 June 16]. <https://www.globenewswire.com/news-release/2025/06/02/3091726/0/en/Interim-Results-from-Ongoing-Phase-2-COURAGE-Trial-Confirm-Potential-to-Improve-the-Quality-of-Semaglutide-GLP-1-receptor-agonist-induced-Weight-Loss-by-Preserving-Lean-Mass.html> (15 September 2025, date last accessed).
 115. Regeneron Phase 2b Study Shows Antibodies Help Preserve Lean Mass During Weight Loss with Semaglutide [Internet]. Patient Care Online, 2025. <https://www.patientcareonline.com/view/regeneron-phase-2b-study-shows-antibodies-help-preserve-lean-mass-during-weight-loss-with-semaglutide> (15 September 2025, date last accessed).
 116. Pistilli EE, Bogdanovich S, Goncalves MD, Ahima RS, Lachey J, Seehra J, et al. Targeting the activin type IIB receptor to improve muscle mass and function in the mdx mouse model of Duchenne muscular dystrophy. *Am J Pathol* 2011;**178**:1287–97. <https://doi.org/10.1016/j.ajpath.2010.11.071>
 117. Attie KM, Allison MJ, McClure T, Boyd IE, Wilson DM, Pearsall AE, et al. A phase 1 study of ACE-536, a regulator of erythroid differentiation, in healthy volunteers. *Am J Hematol* 2014;**89**:766–70. <https://doi.org/10.1002/ajh.23732>
 118. Cadena SM, Tomkinson KN, Monnell TE, Spaitis MS, Kumar R, Underwood KW, et al. Administration of a soluble activin type IIB receptor promotes skeletal muscle growth independent of fiber type. *J Appl Physiol* (1985). 2010;**109**:635–42. <https://doi.org/10.1152/jappphysiol.00866.2009>
 119. Nathan R, Materna C, Welch D, Nurse T, Lema E, Gudelsky A, et al. P.191 RKER-065, a novel muscle and bone anabolic, increased muscle, grip strength and trabecular bone in a mouse model of Duchenne muscular dystrophy. *Neuromuscul Disord* 2022;**32**:S125. <https://doi.org/10.1016/j.nmd.2022.07.346>
 120. Keros Therapeutics Announces Initial Topline Results From the Phase 1 Clinical Trial of KER-065 In Healthy Volunteers [press release]. BioSpace; 2025 April 1. <https://www.biospace.com/press-releases/keros-therapeutics-announces-initial-topline-results-from-the-phase-1-clinical-trial-of-ker-065-in-healthy-volunteers> (15 September 2025, date last accessed).
 121. Keros Therapeutics. (Keros Therapeutics to Develop KER-065 for the Treatment of Obesity) [press release]. Lexington, MA: Keros Therapeutics, Inc., 2024. <https://ir.kerostx.com/news-releases/news-release-details/keros-therapeutics-develop-ker-065-treatment-obesity> (15 September 2025, date last accessed).
 122. Schang G, de Molliens MP, Brûlé E, Chauvet C, Denis JF, Sours A, et al. FRI017 precision-engineered activin and GDF ligand trap HS235: a novel lean mass-preserving treatment for obesity. *J Endocr Soc* 2023;**7**:bvad114.028. <https://doi.org/10.1210/jendso/bvad114.028>
 123. 35Pharma Inc. (35Pharma Announces Oral Presentation Demonstrating Fat-Selective Weight Loss with HS235 at Obesity Week 2023) [press release]. Montreal (QC): 35Pharma Inc., 2023. <https://35pharma.com/35pharma-announces-oral-presentation-demonstrating-fat-selective-weight-loss-with-hs235-at-obesity-week-2023/>. (15 September 2025, date last accessed).
 124. Melson E, Ashraf U, Papamargaritis D, Davies MJ. What is the pipeline for future medications for obesity? *Int J Obes (Lond)* 2025;**49**:433–51. <https://doi.org/10.1038/s41366-024-01473-y>
 125. Rooks D, Petricoul O, Praestgaard J, Bartlett M, Laurent D, Roubenoff R. Safety and pharmacokinetics of bimagrumb in healthy older and obese adults with body composition changes in the older cohort. *J Cachexia Sarcopenia Muscle* 2020;**11**:1525–34. <https://doi.org/10.1002/jcsm.12639>
 126. Barrett D, Bilic S, Chung Y, Cote SM, Iarrobino R, Kacena K, et al. A randomized phase 1 safety, pharmacokinetic and pharmacodynamic study of the novel myostatin inhibitor apitegromab (SRK-015): a potential treatment for spinal muscular atrophy. *Adv Ther* 2021;**38**:3203–22. <https://doi.org/10.1007/s12325-021-01757-z>
 127. Muramatsu H, Kuramochi T, Katada H, Ueyama A, Ruike Y, Ohmine K, et al. Novel myostatin-specific antibody enhances muscle strength in muscle disease models. *Sci Rep* 2021;**11**:2160. <https://doi.org/10.1038/s41598-021-81669-8>
 128. Guo T, Jou W, Chanturiya T, Portas J, Gavrilova O, McPherron AC. Myostatin inhibition in muscle, but not adipose tissue, decreases fat mass and improves insulin sensitivity. *PLoS One* 2009;**4**:e4937. <https://doi.org/10.1371/journal.pone.0004937>
 129. Muntoni F, Byrne BJ, McMillan HJ, Ryan MM, Wong BL, Dukart J, et al. The clinical development of taldefgrobep Alfa: an anti-myostatin adnectin for the treatment of duchenne muscular dystrophy. *Neurol Ther* 2024;**13**:183–219. <https://doi.org/10.1007/s40120-023-00570-w>
 130. Gonzalez Trotter D, Donahue S, Wynne C, Ali S, Parasoglou P, Boyapati A, et al. GDF8 and activin A are the key negative regulators of muscle mass in postmenopausal females: a randomized phase I trial. *Nat Commun* 2025;**16**:4376. <https://doi.org/10.1038/s41467-025-59380-3>

131. Hatzimichael E, Timotheatou D, Koumpis E, Benetatos L, Makis A. Luspatercept: a new tool for the treatment of anemia related to β -thalassemia, myelodysplastic syndromes and primary myelofibrosis. *Diseases* 2022;**10**:85. <https://doi.org/10.3390/diseases10040085>
132. Raje N, Vallet S. Sotatercept, a soluble activin receptor type 2A IgG-Fc fusion protein for the treatment of anemia and bone loss. *Curr Opin Mol Ther* 2010;**12**:586–97.
133. Carrancio S, Markovics J, Wong P, Leisten J, Castiglioni P, Groza MC, et al. An activin receptor IIA ligand trap promotes erythropoiesis resulting in a rapid induction of red blood cells and haemoglobin. *Br J Haematol* 2014;**165**: 870–82. <https://doi.org/10.1111/bjh.12838>
134. Chen MM, Zhao YP, Zhao Y, Deng SL, Yu K. Regulation of myostatin on the growth and development of skeletal muscle. *Front Cell Dev Biol* 2021;**9**: 785712. <https://doi.org/10.3389/fcell.2021.785712>
135. Patel B, Moosavi L. Luspatercept [Updated 2024 Oct 14]. In *StatPearls* [Internet]. Treasure Island (FL): StatPearls Publishing, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK560635/> (1 September 2025, date last accessed).
136. Bhasin S, Jasuja R. Selective androgen receptor modulators as function promoting therapies. *Curr Opin Clin Nutr Metab Care* 2009;**12**:232–40. <https://doi.org/10.1097/MCO.0b013e32832a3d79>
137. Narayanan R, Coss CC, Dalton JT. Development of selective androgen receptor modulators (SARMs). *Mol Cell Endocrinol* 2018;**465**:134–42. <https://doi.org/10.1016/j.mce.2017.06.013>
138. Yoon MS. mTOR as a key regulator in maintaining skeletal muscle mass. *Front Physiol* 2017;**8**:788. <https://doi.org/10.3389/fphys.2017.00788>
139. Hamlaoui K. Selective Androgen Receptor Modulators (SARMs). *Best Pract Res Clin Endocrinol Metab* 2022;**36**:101686. <https://doi.org/10.33552/ARAR.2022.02.000540>
140. Gao W, Reiser PJ, Coss CC, Phelps MA, Kearbey JD, Miller DD, et al. Selective androgen receptor modulator treatment improves muscle strength and body composition and prevents bone loss in orchidectomized rats. *Endocrinology* 2005;**146**:4887–97. <https://doi.org/10.1210/en.2005-0572>
141. Veru Inc. (Veru Reports Muscle Data From 5 Clinical Studies of Enobosarm That Support the Advancement of Enobosarm in Combination with Weight-Loss GLP-1 Drugs, Ozempic®, Wegovy®, or Mounjaro®, to Optimize Weight Loss by Preventing Muscle Wasting and Further Increasing Fat Loss in a Phase 2b Obesity Clinical Study). [Internet]. Miami (FL): GlobeNewswire, 2023. <https://ir.verupharma.com/news-events/press-releases/detail/189/veru-reports-muscle-data-from-5-clinical-studies-of> (1 August 2025, date last accessed).
142. Veru Inc. (Cardiometabolic Obesity Drug Program: Enobosarm Clinical Data). [Internet]. Miami (FL): Veru Inc., 2025. <https://verupharma.com/pipeline/cardiometabolic-obesity-drug-program/#enobosarm-data> (1 August 2025, date last accessed).
143. Veru Inc. Veru Reports Positive Results From Phase 2b QUALITY and Maintenance Extension Study Showing Enobosarm Significantly Reduced Body Weight Regain, Prevented Fat Regain and Preserved Lean Mass After Semaglutide Discontinuation [Internet]. Miami (FL): Veru Inc., Yahoo! Finance, 2025. <https://finance.yahoo.com/news/veru-reports-positive-results-phase-123000078.html?guccounter=1> (1 August 2025, date last accessed).
144. Basaria S, Collins L, Dillon EL, Orwoll K, Storer TW, Miciek R, et al. The safety, pharmacokinetics, and effects of LGD-4033, a novel nonsteroidal oral, selective androgen receptor modulator, in healthy young men. *J Gerontol A Biol Sci Med Sci* 2013;**68**:87–95. <https://doi.org/10.1093/gerona/gls078>
145. Viking Therapeutics Inc. (VK5211—Selective Androgen Receptor Modulator in Development for Hip Fracture and Muscle Wasting) [Internet]. San Diego (CA): Viking Therapeutics Inc., 2025. <https://vikingtherapeutics.com/pipeline/other-pipeline-programs/vk5211/> (1 August 2025, date last accessed).
146. Miller CP, Shomali M, Lyttle CR, O'Dea LS, Herendeen H, Gallacher K, et al. Design, synthesis, and preclinical characterization of the selective androgen receptor modulator (SARM) RAD140. *ACS Med Chem Lett* 2011;**2**:124–9. <https://doi.org/10.1021/ml1002508>
147. Neil D, Clark RV, Magee M, Billiard J, Chan A, Xue Z, et al. GSK2881078, a SARM, produces dose-dependent increases in lean mass in healthy older men and women. *J Clin Endocrinol Metab* 2018;**103**:3215–24. <https://doi.org/10.1210/clinem.2017-02644>
148. Mohan D, Rossiter H, Watz H, Fogarty C, Evans RA, Man W, et al. Selective androgen receptor modulation for muscle weakness in chronic obstructive pulmonary disease: a randomised control trial. *Thorax* 2023;**78**:258–66. <https://doi.org/10.1136/thorax-2021-218360>
149. Unlocking the Power of RAD-140 Testosterone: Benefits, Dosage, and Side-Effects [Internet]. Swolverine. <https://swolverine.com/blogs/blog/rad-140-testosterone-ep0062-benefits-uses-side-effects> (1 August 2025, date last accessed).
150. GeneMedics. RAD-140: Dosage, Benefits & Side Effects [Internet]. Beverly Hills (CA): GeneMedics, 2025. <https://www.genemedics.com/rad-140> (1 August 2025, date last accessed).
151. Misselwitz F, Henderson D, Menakuru SR, Morten E, Roe C, Whitaker G, et al. Pharmacokinetics, pharmacodynamics and bioavailability of ACM-001.1 (S-Pindolol Benzoate) in healthy volunteers. *J Cachexia Sarcopenia Muscle* 2025;**16**:e13651. <https://doi.org/10.1002/jcsm.13651>
152. Pötsch MS, Tschirner A, Palus S, von Haehling S, Doehner W, Beadle J, et al. The anabolic catabolic transforming agent (ACTA) espidolol increases muscle mass and decreases fat mass in old rats. *J Cachexia Sarcopenia Muscle* 2014;**5**:149–58. <https://doi.org/10.1007/s13539-013-0125-7>
153. Actimed Therapeutics. Actimed Therapeutics Enrolls First Patients in Phase 2a PROACT Study Programme to Assess S-Pindolol Benzoate in Obese Patients During and Post-GLP-1 Agonist Therapy [Internet]. London: Actimed Therapeutics, 2025. <https://actimedtherapeutics.com/news/38/54/Actimed-Therapeutics-Enrolls-First-Patients-in-Phase-2a-PROACT-Study-Programme-to-Assess-S-pindolol-benzoate-in-Obese-Patients-During-and-Post-GLP-1-Agonist-Therapy.html> (1 August 2025, date last accessed).
154. Sardeli AV, Komatsu TR, Mori MA, Gáspari AF, Chacon-Mikahil MPT. Resistance training prevents muscle loss induced by caloric restriction in obese elderly individuals: a systematic review and meta-analysis. *Nutrients* 2018;**10**:423. <https://doi.org/10.3390/nu10040423>
155. Ralston GW, Kilgore L, Wyatt FB, Buchan D, Baker JS. Weekly training frequency effects on strength gain: a meta-analysis. *Sports Med Open* 2018;**4**: 36. <https://doi.org/10.1186/s40798-018-0149-9>
156. Farsijani S, Cauley JA, Santanasto AJ, Glynn NW, Boudreau RM, Newman AB. Transition to a more even distribution of daily protein intake is associated with enhanced fat loss during a hypocaloric and physical activity intervention in obese older adults. *J Nutr Health Aging* 2020;**24**:210–7. <https://doi.org/10.1007/s12603-020-1313-8>
157. Anker SD, Ji L, Kindel T, Coats AJS, Ojji D, Puente Barragán A, et al. iCARDIO Alliance Global Implementation Guidelines for the Management of Obesity 2025—Focus on prevention and treatment of cardiometabolic disease. *Global Cardiology* 2025;**3**:181–206. <https://doi.org/10.4081/cardio.2025.86>
158. Chopra V, Khan MS, Abdelhamid M, Abraham WT, Amir O, Anker SD, et al. iCARDIO alliance global implementation guidelines on heart failure 2025. *Global Cardiology* 2025;**3**:35–62. <https://doi.org/10.4081/cardio.2025.70>
159. Holeček M. Beta-hydroxy-beta-methylbutyrate supplementation and skeletal muscle in healthy and muscle-wasting conditions. *J Cachexia Sarcopenia Muscle* 2017;**8**:529–41. <https://doi.org/10.1002/jcsm.12208>
160. Desai I, Pandit A, Smith-Ryan AE, Simar D, Candow DG, Kaakoush NO, et al. The effect of creatine supplementation on lean body mass with and without resistance training. *Nutrients* 2025;**17**:1081. <https://doi.org/10.3390/nu17061081>
161. Amare F, Alemu Y, Enichalew M, Demilie Y, Adamu S. Effects of aerobic, resistance, and combined exercise training on body fat and glucolipid metabolism in inactive middle-aged adults with overweight or obesity: a randomized trial. *BMC Sports Sci Med Rehabil* 2024;**16**:189. <https://doi.org/10.1186/s13102-024-00982-7>
162. Bai H. Nutrition and hypertrophy fueling muscle growth. *J Cell Mol Pharmacol* 2023;**7**:181.
163. Kato M, Ono S, Seko H, Nakatani E, Seo Y, Omote T, et al. Safety and efficacy of home-based neuromuscular electrical stimulation on physical function in frail older patients with heart failure: a randomized crossover study. *Eur J Prev Cardiol* 2024;**31**:zwae175.157. <https://doi.org/10.1093/eurjpc/zwae175.157>
164. McLeod C, Norman R, Litton E, Saville BR, Webb S, Snelling TL. Choosing primary endpoints for clinical trials of health care interventions. *Contemp Clin Trials Commun* 2019;**16**:100486. <https://doi.org/10.1016/j.conctc.2019.100486>
165. Zieroth S, Pinto F, Anker SD, Abraham WT, Atherton JJ, Butler J, et al. Applicability of obesity clinical practice guidelines in low- and middle-income countries. *Global Cardiology* 2025;**3**:96–106. <https://doi.org/10.4081/cardio.2025.77>
166. Millstein RA. Measuring outcomes in adult weight loss studies that include diet and physical activity: a systematic review. *J Nutr Metab* 2014;**2014**: 421423. <https://doi.org/10.1155/2014/421423>
167. Prado CM, Purcell SA, Alish C, Pereira SL, Deutz NE, Heyland DK, et al. Implications of low muscle mass across the continuum of care: a narrative review. *Ann Med* 2018;**50**:675–93. <https://doi.org/10.1080/07853890.2018.1511918>